

A Mean-Field Game of Mayoral Endorsements: Theory and Evidence from Brazil and France*

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Abstract

We develop and estimate a structural model of mayoral endorsements in presidential elections as an anonymous Bayesian game with mean-field equilibrium. The mean-field formulation reduces equilibrium computation to a fixed point on the candidate simplex, independent of the number of mayors, making structural estimation tractable at the scale of an entire country — a regime in which existing full-enumeration estimators are computationally infeasible.

Identification follows a BLP-style logit inversion of observed log-odds onto type covariates and aggregate shares. A closed-form two-step OLS estimator complements the asymptotically efficient constrained-MLE alternative; an alternative-specific extension allows per-candidate covariate effects. Monte Carlo at Brazilian scale (5,570 municipalities) confirms $\sqrt{n}$ consistency and near-nominal coverage under structural-delta standard errors.

We apply the framework to two contrasting institutional settings. In Brazil’s coalition-mediated endorsement system (2014, 5,623 mayors), the structural prize parameter is precisely estimated at $\hat{\alpha} \cdot \hat{B} \in [0.56, 0.65]$ (PSD $t > 9$), and bloc fixed effects recover the expected coalition pattern. In France’s individual-parrainage system (2017 and 2022, $\approx 9,500$ mayor sponsors per cycle), the same estimator recovers ideologically-interpretable per-candidate coefficients but rejects a positive prize: $\hat{\alpha} \cdot \hat{B} \approx -0.12$, because the equilibrium share equation cannot reconcile Macron’s 4% mayor-sponsorship share with positive prize-driven sorting. The contrast delineates a clean scope condition: the framework applies to party-mediated systems with prize-maximization incentives, not to individual sincere-signaling systems.

Keywords: mayoral endorsement game, presidential election, anonymous Bayesian game, mean-field equilibrium, structural estimation, BLP inversion, scope conditions, Brazil, France, parrainages

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1 Introduction

Endorsements are a foundational instrument by which political elites coordinate behind candidates and signal information about candidate quality to voters. [Cohen et al. \(2008\)](#) document how invisible-primary endorsements concentrate elite support behind eventual nominees in U.S. presidential races; [Snyder and Ting \(2002\)](#) model parties themselves as informational endorsement institutions; and [Knight and Schiff \(2010\)](#) show that the timing of endorsements during primaries triggers measurable shifts in voter behavior. Voters, in turn, rely on elite cues — of which endorsements are a leading example — as substitutes for full policy knowledge in low-information settings ([Lupia, 1994](#); [Lupia and McCubbins, 1998](#)). In each of these accounts, the individual endorser’s choice depends on what others are doing: endorsement is fundamentally strategic. Yet the structural-empirical literature on endorsements has remained small, largely because the existing estimators of discrete-choice games of incomplete information do not scale to settings with thousands of strategic agents. This paper closes that gap and applies the resulting framework to two contrasting institutional settings — Brazilian coalition-mediated endorsements and French individual parrainages — whose contrast lets us delineate the institutional preconditions under which the prize-driven mean-field mechanism captures observed endorsement behavior.

Brazil is the natural primary setting. With 5,570 municipalities each headed by a directly elected mayor, it offers a strategic-agent population of large size and observable heterogeneity ([Feierherd, 2020](#)). The federal political-budget mechanism — the Fundo de Participação dos Municípios and discretionary infrastructure transfers — generates a quantifiable payoff that aligned mayors capture from winning presidential candidates ([Brollo et al., 2013](#); [Brollo and Nannicini, 2012](#)). A rich literature on Brazilian political accountability ([Ferraz and Finan, 2008, 2011](#); [Bueno, 2018](#); [Zucco Jr, 2013](#)) provides outside-data anchors for the payoff specification, and the cycles of 2014, 2018, and 2022 supply panel variation that separately identifies mayor-side preferences from candidate-side budgets. Crucially, Brazilian endorsements are mediated by political parties: a mayor’s party joining a presidential candidate’s formal coalition transmits an endorsement to the mayor, with party leaders maximizing the coalition-level prize on behalf of their cadres. This is precisely the institutional structure the prize-driven mean-field model is designed to capture.

France’s parrainages system supplies the complementary scope-condition test. Under French electoral law, every authorized elected official must sponsor a presidential candidate personally and publicly, with the list published verbatim by the Conseil constitutionnel. Three institutional differences from Brazil prove decisive: the endorsement decision is individual rather than party-mediated; the 500-signature qualification threshold gives mayors a tactical incentive to back their preferred candidate regardless of winning prospects; and the signal is sincere, ideological, and weakly tied to post-election fiscal allocation. The data thus permit a direct test of where the prize-driven mechanism applies and where ideological

matching dominates.

Estimating a discrete-choice game of incomplete information at the Brazilian scale is, however, computationally infeasible under the prevailing frameworks. The constrained-MLE estimator of [Su and Judd \(2012\)](#), the nested pseudo-likelihood estimator of [Aguirregabiria and Mira \(2007\)](#), and the two-step pseudo-MLE of [Pesendorfer and Schmidt-Dengler \(2008\)](#) all require enumeration of the joint action profile across all I strategic agents — a state space of size $(J+1)^I$ that grows astronomically in I . With $J = 3$ candidates, full enumeration over 5,570 mayors involves more than $10^{3,300}$ profiles. Set-identification approaches under multiple equilibria ([Tamer, 2003](#); [Ciliberto and Tamer, 2009](#); [Bajari et al., 2010](#); [de Paula and Tang, 2012](#)) address identification under multiplicity but inherit the same enumeration cost. Existing applications of incomplete-information discrete-choice games — including the small-game implementation of the endorsement model in [Vitorino \(2012\)](#) and [Su \(2014\)](#) — accordingly operate at small I . We transport this estimation toolkit to large I by adopting an alternative equilibrium concept: the anonymous-game / mean-field equilibrium framework of [Weintraub et al. \(2008\)](#) and [Adlakha et al. \(2015\)](#), in which agents’ payoffs depend only on aggregate population-weighted action shares and not on the identities of others. Aggregating restores tractability while preserving the strategic content of the model: the equilibrium becomes a fixed point on the candidate simplex, independent of I . The associated identification argument is a BLP-style logit inversion of observed log-odds onto type-level covariates and aggregate shares ([Berry et al., 1995](#); [Nevo, 2000](#); [Berry and Haile, 2014](#)), generalized to incorporate the strategic prize term that characterizes the endorsement game.

This paper makes three contributions. *Methodologically*, we formulate the mayoral endorsement game as a static, anonymous Bayesian game with mean-field equilibrium, prove existence by Brouwer’s theorem, and provide a sufficient condition for uniqueness in the empirically relevant parameter region. The framework reduces equilibrium computation from $O((J+1)^I)$ to a J -dimensional fixed point. Identification proceeds via BLP-style logit inversion, yielding a closed-form two-step OLS estimator complemented by the asymptotically efficient constrained-MLE alternative of [Su and Judd \(2012\)](#) and an alternative-specific extension for settings with strong ideological matching. *Numerically*, Monte Carlo at the Brazilian scale ($K = 30$ types, $n = 5,570$ mayors) demonstrates $\sqrt{n}$ consistency with sub-millisecond runtime, and the structural-delta variance of [Pesendorfer and Schmidt-Dengler \(2008\)](#) achieves near-nominal CI coverage. *Substantively*, the contrast between Brazil and France delineates the model’s institutional scope. The Brazilian application (2014 cycle, coalition-based proxy) recovers a precise structural-prize estimate $\hat{\alpha} \cdot \hat{B} \in [0.56, 0.65]$ ($t > 9$) with bloc fixed effects matching the expected coalition pattern; the French application (parrainages 2017 and 2022, $\approx 9,500$ mayor sponsors per cycle) recovers ideologically-interpretable per-candidate coefficients but rejects a positive prize, with $\hat{\alpha} \cdot \hat{B} \approx -0.12$.

The contrast pins down when the prize-driven mean-field mechanism captures endorsement (party-mediated systems) and when it does not (individual sincere-signaling systems with ballot-qualification motives).

The rest of this article is organized as follows. Section 2 formulates the endorsement game as an anonymous Bayesian game with mean-field equilibrium and establishes existence. Section 3 develops the inverse-mapping identification argument in the spirit of [Berry et al. \(1995\)](#). Section 4 presents a parametric instance of the model that fixes notation for what follows. Section 5 discusses both a closed-form two-step OLS estimator and the constrained-MLE estimator of [Su and Judd \(2012\)](#), and introduces an alternative-specific extension that allows per-candidate covariate effects. Section 6 reports a Monte Carlo study at the empirical scale of Brazil ($K = 30$ mayor types, $n = 5,570$ mayors) verifying $\sqrt{n}$ -rate consistency. Section 7 estimates the model on Brazil’s 2014 cycle using a coalition-based proxy and reports a Brollo-style external anchor for B . Section 8 applies the alternative-specific estimator to French parrainages from 2017 and 2022 and uses the contrast with Brazil to delineate the model’s institutional scope. Section 9 concludes.

2 The Model

We model mayoral endorsement decisions as a static, simultaneous Bayesian game with incomplete information and a continuum-of-types representation in the spirit of [Weintraub et al. \(2008\)](#) and [Adlakha et al. \(2015\)](#). Each mayor chooses one endorsement (or none) given private preference shocks; payoffs depend on observable mayor characteristics, the identity of the winning candidate, and the size of the budgetary prize controlled by that winner. The defining departure from prior empirical implementations of the endorsement game — and the source of all subsequent tractability gains — is that payoffs depend on others’ actions only through aggregate endorsement shares, not through the identities of individual peers. This anonymity assumption converts the equilibrium fixed point from a $(J+1)^I$ -dimensional object into a J -dimensional one, where I is the number of mayors and J the number of presidential candidates. Estimation, identification, and computation simplify accordingly.

2.1 Players, types, and actions

There are I mayors, indexed by $i \in \mathcal{I} = \{1, \dots, I\}$, and J presidential candidates indexed by $j \in \mathcal{J}_+ = \{1, \dots, J\}$. Each mayor either endorses one candidate or makes no public endorsement, so the action set is $\mathcal{J} = \{0\} \cup \mathcal{J}_+$ with $j = 0$ denoting non-endorsement. We write $e_i \in \mathcal{J}$ for mayor i ’s action and $\mathbf{e} = (e_1, \dots, e_I) \in \mathcal{J}^I$ for the action profile.

Mayors are heterogeneous in observable characteristics. Let $\tau_i \in \mathcal{T}$ denote mayor i ’s type, where $\mathcal{T}$ is a finite set with $|\mathcal{T}| = K$. The type vector $X(\tau) \in \mathbb{R}^{n \times X}$ collects ob-

servable mayor and municipality covariates. Empirically, types index combinations of region, city size, and party affiliation (see Section 7). The type distribution $f(\tau)$ is observable; let $w(\tau)$ be a weight reflecting type τ 's share of the national presidential electorate, so $\sum_{\tau} f(\tau)w(\tau) = 1$. Each mayor also draws an idiosyncratic preference shock vector $\varepsilon_i = (\varepsilon_i(0), \varepsilon_i(1), \dots, \varepsilon_i(J))$, observed only by mayor i , with components iid Type I extreme value across (i, j) .

2.2 Payoffs and aggregate shares

Let $s = (s_0, s_1, \dots, s_J) \in \Delta^J$ denote the population-weighted vector of endorsement shares, i.e.,

$$s_j = \sum_{\tau \in \mathcal{T}} f(\tau) w(\tau) \sigma(j | \tau), \quad (1)$$

where $\sigma(j | \tau)$ is the type-conditional probability that a mayor of type τ endorses candidate j . The vector s is the only feature of the action profile that enters payoffs; in this sense the game is *anonymous* (Jensen, 2010; Acemoglu and Jensen, 2013).

Candidate j wins iff her endorsement share is maximal,

$$W(s) = \arg \max_{j \in \mathcal{J}_+} s_j, \quad (2)$$

which is generically a singleton in the parameter region of empirical interest; ties form a measure-zero subset and are handled by convention (e.g., uniform tiebreak) in the few parameterizations where they arise.

Mayor i 's deterministic payoff from action $e_i = j$ when the share vector is s takes the form

$$\pi(j, s, \tau_i; \theta) = \begin{cases} X(\tau_i)' \beta + \alpha(\tau_i) \frac{B_j}{s_j} & \text{if } j \in W(s), \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

Here $\beta \in \mathbb{R}^{n_x}$ are common preference weights on observable characteristics, $\alpha(\tau)$ is the type-specific marginal utility of money (heterogeneity in fiscal needs, governance autonomy, or political ideology), and B_j is the budgetary prize controlled by candidate j . The denominator s_j replaces the small-game expression “budget divided by number of endorsers”: in the large- I limit, the total prize $B_j \cdot I$ is divided among the $I s_j$ mayors who endorsed j , yielding a per-endorser amount of B_j/s_j that remains finite and incentive-relevant.¹ Mayors who do not endorse, or who endorse a candidate that does not win, receive zero deterministic payoff. The full random utility is $u_i(j, s, \tau_i, \varepsilon_i; \theta) = \pi(j, s, \tau_i; \theta) + \varepsilon_i(j)$.

¹This rescaling is the natural one in aggregative-game representations of large electorates; see Bodoh-Creed (2013) and Adlakha et al. (2015). It also has direct empirical support: total federal discretionary transfers (FPM and analogous instruments) scale roughly proportionally with the number of municipalities served, so the per-municipality alignment premium does not vanish as the country grows.

2.3 Beliefs and the type-conditional best response

Each mayor's information set comprises (i) the type distribution f and weights w , (ii) common parameters θ , and (iii) the mayor's own type τ_i and shock ε_i . Mayors form rational beliefs about other mayors' actions through the equilibrium share s . Given s , the expected utility of action j conditional on type τ is

$$U(j, s, \tau, \varepsilon; \theta) = \pi(j, s, \tau; \theta) + \varepsilon(j), \quad (4)$$

where the absence of a sum over e_{-i} (cf. the small-game formulation in earlier work) is the substantive content of the anonymity assumption: only the aggregate s enters, and s is taken as common knowledge in equilibrium.

Under the Type I extreme value assumption (McFadden, 1974; Train, 2009), the type-conditional best-response probability is

$$\sigma(j \mid \tau) = \frac{\exp(\pi(j, s, \tau; \theta))}{1 + \sum_{k \in \mathcal{J}_+} \exp(\pi(k, s, \tau; \theta))}, \quad j \in \mathcal{J}_+, \quad (5)$$

with the outside option normalized to $\pi(0, s, \tau; \theta) \equiv 0$ and $\sigma(0 \mid \tau) = 1 - \sum_j \sigma(j \mid \tau)$.

2.4 Mean-field equilibrium

Definition 1 (Mean-field equilibrium). A pair $(\sigma^*, s^*) \in \mathbb{R}_+^{K(J+1)} \times \Delta^J$ is a *mean-field equilibrium* if

[(i)]

1. $\sigma^*(j \mid \tau)$ satisfies the best-response equation (5) for every (τ, j) given $s = s^*$, and
2. s^* satisfies the consistency condition (1) given $\sigma = \sigma^*$.

Proposition 1 (Existence). *A mean-field equilibrium exists. The mapping $\Phi : \Delta^J \rightarrow \Delta^J$ defined by combining (1) and (5) is a continuous self-map of the compact, convex simplex Δ^J and therefore admits a fixed point by Brouwer's theorem.*

The proof, given in Appendix B, is standard. Uniqueness is not guaranteed and we take this seriously in identification (Section 3) and estimation (Section 5). When uniqueness fails we report the equilibrium-correspondence-based identified set in the spirit of Tamer (2003) and Ciliberto and Tamer (2009).

2.5 Approximation of the finite- I Bayesian game

Definition 1 characterizes the limit object as $I \rightarrow \infty$. For empirical work with finite I , we appeal to the approximation result of Bodoh-Creed (2013) for static large mechanisms:

the mean-field equilibrium is an ϵ -equilibrium of the corresponding finite- I Bayesian game with $\epsilon = O(1/\sqrt{I})$ under the regularity assumptions we maintain (continuous payoffs in shares, bounded prize scaling).² For Brazil's $I \approx 5,570$ municipalities, the implied bound is approximately 0.013, an order of magnitude smaller than the standard errors we report; the approximation is therefore innocuous in the empirical setting, and we verify it numerically in our Monte Carlo (Section 6).

3 Identification

The structural parameter vector is $\theta = (\beta, \{\alpha(\tau)\}_\tau, \{B_j\}_j)$. We discuss the identification of each component in turn, anchored by an inverse-mapping argument analogous to that of [Berry et al. \(1995\)](#) and [Berry and Haile \(2014\)](#).

3.1 Inverse mapping from shares to log-odds

For any mean-field equilibrium (σ^*, s^*) , dividing the best-response equation (5) by the no-endorsement probability and taking logs yields the type-by-candidate identity

$$\log \frac{\sigma^*(j | \tau)}{\sigma^*(0 | \tau)} = \pi(j, s^*, \tau; \theta) = X(\tau)' \beta + \alpha(\tau) \cdot \frac{B_j}{s_j^*} \mathbf{1}\{j \in W(s^*)\}, \quad (6)$$

for every $\tau \in \mathcal{T}$ and $j \in \mathcal{J}_+$. The left-hand side is observable from the data once we estimate type-conditional choice probabilities by their empirical analogs $\hat{\sigma}(j | \tau) = \hat{N}_{\tau,j} / \hat{N}_\tau$, where $\hat{N}_\tau$ is the number of mayors of type τ in the sample and $\hat{N}_{\tau,j}$ the count endorsing j . The aggregate share s_j^* is similarly observable from the population-weighted endorsement frequency.

The right-hand side of (6) has a knife-edge structure: the prize term enters only on the indicator $\mathbf{1}\{j \in W(s^*)\}$, so for non-winning candidates $j \notin W(s^*)$ the equation collapses to $\log[\sigma^*(j | \tau) / \sigma^*(0 | \tau)] = X(\tau)' \beta$. With β shared across j and the outside option normalized to $\pi(0, \cdot) = 0$, this is the same equation for every losing candidate j . The model therefore predicts $\sigma^*(j | \tau) = \sigma^*(0 | \tau)$ for all losing j , which makes the loser equations *informationally redundant* for θ identification: only the K winner equations identify (β, α, B) within a cycle. The loser equations remain useful as overidentifying restrictions: the testable model implication $\hat{\sigma}(\text{loser} | \tau) \approx \hat{\sigma}(0 | \tau)$ is a specification check, and section 7.2 reports the empirical analog.

²The Bodoh-Creed (2012) bound is for static settings and applies directly here. The dynamic analog is [Weintraub et al. \(2008\)](#) and [Adlakha et al. \(2015\)](#), which we cite for the broader programme although those bounds require additional regularity (continuous state, bounded payoffs) that our B_j/s_j specification with boundary blow-up does not satisfy without the lower-bounded restriction $s_j \geq \underline{s}$ developed in Appendix B.2.

3.2 Identifying β and α within a single cycle

In a single presidential cycle, the unique winner $j^* = W(s^*)$ delivers K informative log-odds equations indexed by τ :

$$\log \frac{\sigma^*(j^* | \tau)}{\sigma^*(0 | \tau)} = X(\tau)' \beta + \alpha(\tau) \cdot \frac{B_{j^*}}{s_{j^*}^*}. \quad (7)$$

The unknowns on the right are n_X components of β , the K type-specific marginal utilities $\alpha(\tau)$, and the J candidate prizes B_j . Within a single cycle we observe the winner only, so $\{B_j : j \neq j^*\}$ are not pinned by (7). We restrict the within-cycle estimand to (β, α, B_{j^*}) . Furthermore, $\alpha(\tau) \cdot B_{j^*}$ enters multiplicatively, so only their product is identified per τ (Section 3.3 addresses the cross-cycle separation). The order condition for joint identification of $(\beta, \{\alpha(\tau) \cdot B_{j^*}\}_\tau)$ from the K winner equations is

$$K \geq n_X + K \quad \text{if } \alpha(\tau) \text{ is type-specific,} \quad (8)$$

which fails (the $\alpha(\tau)$ are perfectly multicollinear with type fixed effects in the cross-section). Identification accordingly requires a parametric restriction on $\alpha(\tau)$, the simplest being $\alpha(\tau) \equiv \alpha$ common across types — which we adopt in the empirical exercise of Section 7. Under this restriction, the order condition becomes $K \geq n_X + 1$, comfortably satisfied for $K \geq 26$. Heterogeneity in $\alpha(\tau)$ is recovered through $X(\tau)$ -driven variation: parametrising $\alpha(\tau) = \alpha_0 + \alpha_1 \cdot z(\tau)$ for a covariate z excluded from β adds one identified parameter per included z . Identification of β versus the α component requires the standard exclusion restriction — at least one component of $X(\tau)$ that does not also enter α .

3.3 Separating α from B via cross-cycle variation

Within a single cycle, $\alpha(\tau)$ and B_j enter (6) multiplicatively as $\alpha(\tau) \cdot B_j$, so only the product is identified per (τ, j) . Two complementary strategies break this multiplicative ambiguity.

Cross-cycle identification. Indexing cycles $c \in \{1, \dots, C\}$, candidates and budgetary prizes vary across cycles while mayor-type marginal utilities are arguably stable:

$$\log \frac{\sigma_c^*(j | \tau)}{\sigma_c^*(0 | \tau)} = X(\tau)' \beta + \alpha(\tau) \cdot \frac{B_{j,c}}{s_{j,c}^*} \mathbf{1}\{j \in W(s_c^*)\}. \quad (9)$$

With $C \geq 2$ presidential cycles in the data and overlapping types across cycles, $\alpha(\tau)$ is identified by within-type cross-cycle variation in candidate-cycle pairs, while $\{B_{j,c}\}$ are identified by within-cycle cross-type variation.

Anchoring B_j from outside data. An alternative is to fix $B_{j,c}$ from realized post-election fiscal-alignment premia, following [Brollo et al. \(2013\)](#). Concretely, one regresses post-election federal transfers (FPM, FUNDEB) to municipality τ on an indicator that τ 's mayor was aligned with the cycle- c winner, recovering an empirical alignment premium that proxies for $B_{j,c}$. With $\hat{B}_{j,c}$ in hand, only $(\beta, \{\alpha(\tau)\})$ remain to be estimated, which is a linear regression. We adopt this strategy in our preferred specification and present the cross-cycle-only specification as a robustness check.

3.4 Identification under multiple equilibria

The fixed point in Definition 1 need not be unique. When uniqueness fails, the parameter vector θ is partially identified: every θ for which *some* equilibrium $s^*(\theta)$ is consistent with the data is admissible ([Tamer, 2003](#); [Ciliberto and Tamer, 2009](#); [de Paula and Tang, 2012](#)). We follow two complementary defenses. First, in the Monte Carlo we verify that the parametric region of interest exhibits a unique equilibrium (the dominant case in our calibration). Second, when applying to data we report the identified set under multiplicity by enumerating equilibria via best-response iteration from a grid of starting points, and intersecting the parameter sets that rationalize each.

3.5 Discussion: how the framework relates to BLP

Equation (6) is the discrete-choice game analog of the BLP demand inversion ([Berry et al., 1995](#); [Nevo, 2000](#)): log-odds of choice equal a linear-in-parameters function of observables and an aggregate sufficient statistic. The differences are (i) the "instrument" for the endogenous aggregate s here is provided by the equilibrium fixed point rather than by cost-side instruments, (ii) heterogeneity is at the mayor-type level rather than at the consumer level, and (iii) the prize term B_j/s_j introduces a non-linearity in s that is absent in standard BLP. We exploit (i) directly through the MPEC formulation in Section 5; the non-linearity (iii) is handled by treating s as an additional optimization variable subject to consistency, in the spirit of [Su and Judd \(2012\)](#) and [Dubé et al. \(2012\)](#).

4 A parametric example

To make the framework concrete and to fix notation for the Monte Carlo of Section 6, we work through a fully specified parametric instance of the model. The instance retains every structural ingredient of Section 2 while reducing notation to a single common α parameter; full type-heterogeneity in $\alpha(\tau)$ is introduced in Section 7.

4.1 Primitives

The example has $K = 30$ mayor types, $J = 3$ presidential candidates plus the outside option, and $n_X = 3$ continuous covariates per type. Type prevalence and population weights are

$$f \sim \text{Dirichlet}(2\mathbf{1}_K), \quad w(\tau) = \frac{\tilde{w}(\tau)}{\sum_{\tau'} f(\tau') \tilde{w}(\tau')}, \quad \tilde{w}(\tau) \stackrel{\text{iid}}{\sim} \text{Lognormal}(0, 0.4), \quad (10)$$

so that $\sum_{\tau} f(\tau)w(\tau) = 1$. Type covariates are

$$X(\tau) \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, I_3), \quad (11)$$

mutually independent across types and components. The parameter vector is $\boldsymbol{\theta} = (\beta, \alpha, B_1, B_2, B_3) \in \mathbb{R}^7$ with $\beta \in \mathbb{R}^3$.

4.2 Payoffs and best response

The deterministic payoff to a mayor of type τ from endorsing candidate j , given equilibrium aggregate share s , is the specialization of equation (3) from Section 2.2:

$$\pi(j, s, \tau; \boldsymbol{\theta}) = \begin{cases} X(\tau)' \beta + \frac{\alpha B_j}{s_j} & \text{if } j = W(s), j > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (12)$$

The Type-I-extreme-value best response then takes the explicit form

$$\sigma(j \mid \tau) = \begin{cases} \frac{\exp(X(\tau)' \beta + \alpha B_{j^*}/s_{j^*})}{Z(\tau)} & \text{if } j = j^* := W(s), \\ \frac{1}{Z(\tau)} & \text{if } j \neq j^*, \end{cases} \quad Z(\tau) = J + \exp(X(\tau)' \beta + \alpha B_{j^*}/s_{j^*}). \quad (13)$$

A direct corollary is the testable model implication that losing-candidate shares equal the no-endorsement share at every type:

$$\sigma(j \mid \tau) = \sigma(0 \mid \tau) \quad \text{for every loser } j \neq j^*, \tau \in \mathcal{T}. \quad (14)$$

This restriction is testable in the data and provides a within-cycle specification check. (Section 7 reports the empirical analog and discusses extensions in which it can be relaxed — most naturally by allowing candidate-specific covariates $X_j(\tau)$ or candidate-specific intercepts δ_j .)

4.3 Mean-field equilibrium and the implicit fixed point

The aggregate share s_{j^*} is determined implicitly by combining the consistency equation (1) with (13) for the winner:

$$s_{j^*} = \sum_{\tau \in \mathcal{T}} f(\tau) w(\tau) \frac{\exp(X(\tau)' \beta + \alpha B_{j^*} / s_{j^*})}{J + \exp(X(\tau)' \beta + \alpha B_{j^*} / s_{j^*})}. \quad (15)$$

Equation (15) is a one-dimensional fixed point in s_{j^*} : given j^* , the right-hand side is monotonic in s_{j^*} and a unique solution exists in $(0, 1)$ under regularity conditions on αB_{j^*} . The other components of s are residually determined by $s_0 = s_j = (1 - s_{j^*})/J$ for $j \neq j^*$.

This is the analog of the BLP demand-aggregation fixed point (Berry et al., 1995): a one-dimensional contraction whose solution can be found by Newton's method or by simple iteration to machine precision in tens of iterations. The Monte Carlo of Section 6 uses damped iteration with damping factor $1/2$, converging in roughly 18 iterations from a uniform initial point.

4.4 Reading the example

It is useful to track which structural object is in play in each role. The covariate term $X(\tau)' \beta$ governs the propensity of a type- τ mayor to endorse a winner conditional on a winner being present, distinguishing types of mayor in their willingness to endorse rather than abstain. The prize term $\alpha B_{j^*} / s_{j^*}$ governs the additional pull of the winning candidate; it interacts with the equilibrium share through $1/s_{j^*}$, so the prize per endorser is endogenously determined by who else is endorsing.

The interpretation of α is the type- τ -mayor's marginal utility of one unit of budget. In single-cycle data, only the product αB_{j^*} is identified; in panel data with at least two cycles, α separates from B via cross-cycle variation in the candidate roster (Section 3.3). The empirical exercise (Section 7) implements both within-cycle and cross-cycle specifications.

5 Estimation

We discuss two estimators, both grounded in the inverse-mapping derivation of Section 3.1. The first is a closed-form two-step OLS estimator analogous to BLP demand inversion (Berry et al., 1995; Nevo, 2000) or to the two-step PML procedure of Pesendorfer and Schmidt-Dengler (2008); it is the workhorse of the Monte Carlo in Section 6. The second is the constrained-MLE estimator of Su and Judd (2012), which jointly imposes the equilibrium constraint at the optimum and is asymptotically efficient.

5.1 Two-step OLS estimator

Step one constructs Laplace-smoothed empirical choice probabilities, empirical type prevalence, and empirical aggregate shares:

$$\hat{\sigma}(j \mid \tau) = \frac{N_{\tau,j} + s}{N_{\tau} + s(J+1)}, \quad s = 0.5 \quad (\text{Jeffreys prior}), \quad (16)$$

$$\hat{f}(\tau) = \frac{N_{\tau}}{n}, \quad N_{\tau} = \sum_j N_{\tau,j}, \quad (17)$$

$$\hat{s}_j = \sum_{\tau} \hat{f}(\tau) w(\tau) \hat{\sigma}(j \mid \tau), \quad (18)$$

$$\hat{j}^* = \arg \max_{j > 0} \hat{s}_j. \quad (19)$$

Step two stacks the inverse-mapping log-odds equations (6) for $j = \hat{j}^*$ across types:

$$\underbrace{\log \frac{\hat{\sigma}(\hat{j}^* \mid \tau)}{\hat{\sigma}(0 \mid \tau)}}_{\equiv y_{\tau}} = X(\tau)' \beta + \frac{\alpha B_{\hat{j}^*}}{\hat{s}_{\hat{j}^*}} + u_{\tau}, \quad (20)$$

where u_{τ} collects sampling noise in $\hat{\sigma}$. With normalization $B_{\hat{j}^*} = 1$, equation (20) is a textbook OLS regression of y_{τ} on $X(\tau)$ and a constant, with K observations and $n_X + 1$ parameters. The coefficient on the constant is $\hat{\alpha}/\hat{s}_{\hat{j}^*}$, from which $\hat{\alpha} = \hat{s}_{\hat{j}^*} \cdot \widehat{\text{coef}}$.

Heteroskedasticity-robust standard errors. Because the dependent variable y_{τ} is a transformation of an empirical proportion, its sampling variance varies systematically across types — larger types (N_{τ} large) have more precise log-odds. We therefore report HC0 (White) standard errors:

$$\hat{V}_{\text{HC0}} = (Z'Z)^{-1} Z' \text{diag}(\hat{u}_{\tau}^2) Z (Z'Z)^{-1}, \quad (21)$$

where $Z = [X, 1/\hat{s}_{\hat{j}^*} \mathbf{1}_K]$ is the regressor matrix.

Asymptotic-correction for first-stage estimation. HC0 standard errors treat $\hat{\sigma}$ as known and consequently underestimate true sampling variability. The full asymptotic standard error follows the construction of [Pesendorfer and Schmidt-Dengler \(2008\)](#) (§3): the second-stage moment is differentiable in the first-stage $\hat{\sigma}$, so the variance of the second-stage estimator includes an additive term capturing first-stage estimation error scaled by the moment Jacobian $\partial m / \partial \hat{\sigma}$. We apply this correction in the empirical exercise (Section 7); the Monte Carlo of Section 6 reports both un-corrected and the implied direction of correction for transparency.

5.2 Constrained-MLE estimator

The two-step estimator is consistent and computationally trivial but inefficient: it does not exploit the equilibrium constraint at the model’s optimum. The constrained-MLE estimator of [Su and Judd \(2012\)](#) jointly maximizes the panel log-likelihood subject to the structural equilibrium and consistency constraints. In the v2 framework, the Mathematical Program with Equilibrium Constraints (MPEC) has the form

$$\max_{\boldsymbol{\theta}, \sigma, s} \sum_{\tau, j} N_{\tau, j} \log \sigma(j \mid \tau) \quad (22)$$

$$\text{s.t. } \sigma(j \mid \tau) = \frac{\exp(\pi(j, s, \tau; \boldsymbol{\theta}))}{\sum_k \exp(\pi(k, s, \tau; \boldsymbol{\theta}))} \quad \forall (\tau, j), \quad (23)$$

$$s_j = \sum_{\tau} f(\tau) w(\tau) \sigma(j \mid \tau) \quad \forall j, \quad (24)$$

$$\sum_j \sigma(j \mid \tau) = 1 \quad \forall \tau. \quad (25)$$

Decision variables include the structural parameters $\boldsymbol{\theta}$, the type-conditional probabilities σ , and the aggregate shares s . With $K = 30$, $J = 3$, $n_X = 3$ this is a problem with roughly 40 structural variables, 123 choice-probability and share variables, and 123 equality constraints — well within the reach of modern interior-point solvers ([Wächter and Biegler, 2006](#)). Solver convergence is typically reached in tens of iterations from a two-step warm start.

The constrained-MLE estimator delivers (i) asymptotically efficient point estimates, (ii) Lagrange multipliers from which analytic standard errors follow via the implicit function theorem and the multiplier-augmented Hessian, and (iii) a likelihood ratio statistic for nested-model testing ([Egesdal et al., 2015](#); [Dubé et al., 2012](#)). In the empirical exercise we use the two-step estimator as a warm-started initial guess and the constrained-MLE as the final reported estimator.

5.3 Alternative-specific extension

The two-step OLS specification of equation (20) imposes a single coefficient vector β shared across all candidates. This is the appropriate specification when the type covariates $X(\tau)$ capture mayors’ propensity to engage in any prize-seeking endorsement (as in Brazil, where party-mediated coalition strings determine the endorsement signal and the candidate identities matter only through the coalition assignment). When the empirical setting instead exhibits strong ideological matching — different mayor types systematically prefer different candidates on grounds orthogonal to the post-election prize — a single- β specification misattributes ideological cross-sectional variation to the structural prize term, and the prize parameter is poorly recovered.

The natural extension allows each candidate j to have its own coefficient vector β_j on the same $X(\tau)$:

$$\log \frac{\hat{\sigma}(j \mid \tau)}{\hat{\sigma}(0 \mid \tau)} = X(\tau)' \beta_j + \mathbf{1}\{j = \hat{j}^*\} \frac{\alpha B_{\hat{j}^*}}{\hat{s}_{\hat{j}^*}} + u_{\tau,j}, \quad j = 1, \dots, J. \quad (26)$$

This is estimated equation-by-equation by OLS on each candidate's K type-level rows. The winner equation ($j = \hat{j}^*$) carries the additional $1/\hat{s}_{\hat{j}^*}$ regressor and identifies $(\beta_{\hat{j}^*}, \alpha B_{\hat{j}^*})$ jointly; each loser equation identifies β_j alone. Total parameters: $J \cdot n_X + 1$. Standard errors are computed equation-by-equation under HC0/HC1/HC2/HC3 or, for the winner equation, the structural-delta variance of [Pesendorfer and Schmidt-Dengler \(2008\)](#) with multinomial-margin variance $1/[\hat{\sigma}(\tau, j)\hat{\sigma}(\tau, 0)N_\tau]$. Implementation is at `code/python/mc_v2/estimator_v2.py` (`two_step_alt_specific`).

The alternative-specific extension nests the original two-step specification as the restriction $\beta_j = \beta$ for all j . A Wald or Lagrange-multiplier test of this restriction is informative about whether the data exhibit candidate-specific ideological matching. Section 8 applies the alternative-specific estimator to French parrainages, where the per-candidate β_j recovery is the key to interpreting the structural-prize result.

5.4 Equilibrium selection

When multiple equilibria coexist, the question of *which* equilibrium is being played in the data is substantive. We follow [Bajari et al. \(2010\)](#) in adopting a single-equilibrium-selection rule based on the observed data: among equilibria $(\sigma^{(\ell)}, s^{(\ell)})$ admissible at θ , we select the one with shares closest to the empirical $\hat{s}$. Concretely, the structural model is identified from

$$\theta \in \arg \max_{\theta} \log \text{lik}(\theta, \sigma^{*(\theta)}, s^{*(\theta)}), \quad \text{where} \quad s^{*(\theta)} = \arg \min_{s \in \text{MFE}(\theta)} \|s - \hat{s}\|. \quad (27)$$

For the Brazilian application of Section 7, we report robustness checks in which the selection rule is varied (e.g., to the equilibrium with maximum share for the empirically observed winner) and find structural-parameter estimates within sampling error of one another.

5.5 Computation

Both estimators are implemented in Python. The two-step estimator is a single OLS solve and runs in approximately 0.2 ms per replication. The constrained-MLE estimator uses `pyomo` ([Wächter and Biegler, 2006](#)) interfaced with IPOPT and runs in roughly 0.1–0.5 seconds per replication on a single CPU. Both estimators scale gracefully in K and J but are insensitive to I once $\hat{\sigma}$ has been computed in step one. Replication code is available at `code/python/mc_v2/`.

6 Monte Carlo

This section examines the finite-sample performance of the two-step estimator developed in Section 5 on data generated from the model of Section 2. The Monte Carlo is conducted at scales matching the Brazilian application: $K = 30$ mayor types, $J = 3$ presidential candidates, and a sample size of $n = 5,570$ mayors equal to the universe of Brazilian municipalities. We report bias, root mean squared error (RMSE), Monte Carlo standard deviation, and 95% confidence-interval coverage across $R = 500$ independent replications. We additionally study how these statistics scale with the sample size n to verify the standard $\sqrt{n}$ -rate of convergence.

A point of comparison is useful from the outset. The original full-enumeration formulation, which sums over $(J + 1)^I$ joint action profiles, has $4^{5,570} \approx 10^{3,355}$ profiles at the empirical scale — not tractable on any conceivable hardware. The mean-field formulation reduces equilibrium computation to a fixed point on the J -dimensional simplex, independent of I . The Monte Carlo below executes one full estimation in approximately 0.3 milliseconds; $R = 500$ replications complete in under two seconds.

6.1 Data-generating process

We instantiate the model with $K = 30$ mayor types and $n_X = 3$ continuous covariates per type. Type-level covariates $X(\tau)$ are drawn iid standard normal. Type prevalence $f(\tau)$ is drawn from a Dirichlet(2) distribution and population weights $w(\tau)$ are drawn lognormal with location 0 and scale 0.4, then renormalized so that $\sum_{\tau} f(\tau)w(\tau) = 1$. The structural parameters are

$$\begin{aligned}\beta &= (0.5, 0.3, -0.2), \\ \alpha &= 1.5 \quad (\text{common across types in this baseline DGP}), \\ B &= (1.0, 0.4, 0.2).\end{aligned}$$

The B ordering ensures candidate $j = 1$ has the largest budgetary prize and is the unique stable winner under the mean-field equilibrium starting from any uniform initialization. Equilibrium shares satisfy $s_1^* \approx 0.71$ and $s_0^* = s_2^* = s_3^* \approx 0.10$, the latter equality being the model’s testable prediction that losing-candidate shares equal the no-endorsement share.

We solve for the mean-field equilibrium by damped fixed-point iteration on $\Phi : \Delta^J \rightarrow \Delta^J$ defined in Section 2.4, iterating

$$s_{t+1} = \frac{1}{2} s_t + \frac{1}{2} \Phi(s_t)$$

to a tolerance of 10^{-12} in sup-norm. Starting from the uniform initial point, convergence requires 18 iterations. To verify uniqueness within the parameter region of interest, we

re-run the iteration from 20 randomly drawn initial points; all 20 converge to the same equilibrium. (Outside the region of interest, two additional equilibria exist in which $j = 2$ or $j = 3$ wins; these are unstable in the sense that small perturbations of s near them escape toward the $j = 1$ equilibrium.)

For each replication, we draw $n = 5,570$ mayor identities iid from $f(\tau)$, then for each mayor draw an endorsement from $\sigma^*(\cdot \mid \tau)$. The simulated data $\{N_{\tau,j}\}_{\tau,j}$ are tabulated counts of endorsements by type and candidate.

6.2 Estimator

We apply the two-step BLP-style estimator from Section 5. Step one constructs Laplace-smoothed empirical choice probabilities $\hat{\sigma}(j \mid \tau) = (N_{\tau,j} + 0.5)/(N_{\tau} + 0.5(J + 1))$, empirical type prevalence $\hat{f}(\tau)$, and empirical aggregate share $\hat{s}_j$. The empirical winner $\hat{j}^* = \arg \max_{j > 0} \hat{s}_j$ is identified. Step two regresses $\log(\hat{\sigma}(\hat{j}^* \mid \tau)/\hat{\sigma}(0 \mid \tau))$ on $X(\tau)$ and a constant proportional to $1/\hat{s}_{\hat{j}^*}$. Recovering β and the product $\alpha \cdot B_{\hat{j}^*}$ amounts to ordinary least squares with $K = 30$ observations and $n_X + 1 = 4$ regressors. Standard errors are HC0 (White) heteroskedasticity-robust.

Within a single cycle, α and $B_{\hat{j}^*}$ are identified only through their product, so we report estimates of the product and apply the normalization $B_{\hat{j}^*} = 1$ throughout. Cross-cycle separation of α and B requires panel data and is examined in the empirical application (Section 7).

6.3 Results at the empirical scale ($n = 5,570$)

Table 1 summarizes the $R = 500$ replications at the Brazilian sample size. Every replication correctly identifies $j = 1$ as the empirical winner, confirming the unique-equilibrium calibration. The estimator is essentially unbiased at this scale: every bias is below 1% of the true parameter magnitude. Root mean squared errors lie between 0.05 and 0.08 for β and 0.054 for $\alpha B_{\hat{j}^*}$, in line with conventional sample-size expectations for a two-step procedure with $KJ = 90$ moment cells.

Table 1. Monte Carlo summary at $K = 30$, $n = 5,570$, $R = 500$

Parameter	True	Mean estimate	Bias	RMSE	95% Coverage (PSD)
β_1	0.500	0.499	−0.001	0.075	96.6%
β_2	0.300	0.303	0.003	0.069	97.0%
β_3	−0.200	−0.203	−0.003	0.057	97.0%
$\alpha \cdot B_{j^*}$	1.500	1.489	−0.011	0.054	93.6%
Correct winner identification			500/500 (100%)		
Median estimator runtime			0.30 ms / replication		
Total replication runtime			0.2 seconds		

Notes: Standard errors are computed using the Pesendorfer–Schmidt–Dengler structural delta-method (PSD) variance: $\widehat{\text{Var}}(y_\tau) = 1/(\hat{\sigma}(\tau, \hat{j}^*)\hat{\sigma}(\tau, 0)N_\tau)$. CI coverage uses the $\pm 1.96 \times \widehat{\text{SE}}$ Wald interval. PSD slightly over-covers β (96%–97% vs. nominal 95%) due to Laplace-smoothing inflation of $\hat{\sigma}(1-\hat{\sigma})$; HC3 (93.8%–95.8% for β) is at-target but under-covers α at 89%. We adopt PSD as the default to be conservative on α . Section 6.6 compares all five SE variants (HC0/HC1/HC2/HC3/PSD). Computational hardware: laptop CPU (no parallelism, no GPU).

6.4 Convergence rate

To verify the standard $n^{-1/2}$ rate of convergence and the consistency of the estimator, we replicate the experiment at $n \in \{500, 1,000, 2,000, 5,570, 10,000\}$. Table 2 reports RMSE and bias by sample size. The pattern is unambiguous: bias decays toward zero and RMSE shrinks as approximately $n^{-1/2}$, falling by a factor of roughly $\sqrt{20} \approx 4.5$ as n rises from 500 to 10,000.

Table 2. Convergence rate of the two-step estimator

	Sample size n				
	500	1,000	2,000	5,570	10,000
<i>RMSE</i>					
β_1	0.219	0.163	0.122	0.075	0.054
β_2	0.170	0.134	0.108	0.069	0.049
β_3	0.159	0.122	0.096	0.057	0.041
$\alpha \cdot B_{j^*}$	0.266	0.148	0.090	0.054	0.040
<i>Bias</i>					
β_1	-0.156	-0.068	-0.013	-0.001	0.005
β_2	-0.070	-0.032	-0.023	0.003	-0.004
β_3	0.082	0.027	-0.003	-0.003	0.001
$\alpha \cdot B_{j^*}$	-0.238	-0.111	-0.045	-0.011	-0.007

Notes: $R = 500$ replications per cell. Bias is the average estimate minus the true value. RMSE accounts for both bias and variance. The columns correspond to sample sizes from $n = 500$ (a small-sample regime in which the inverse-mapping transformation is inherently nonlinear) up to $n = 10,000$ (slightly larger than Brazil’s municipality count). The $\sqrt{n}$ scaling holds in both bias and RMSE: doubling n approximately reduces both by $1/\sqrt{2}$.

The convergence rate confirms two things. First, the two-step estimator is consistent: bias visibly approaches zero. Second, the small-sample bias at $n = 500$ is non-negligible (up to 30% of β_1) but rapidly attenuates, becoming inconsequential at the empirical sample size. Researchers applying this framework to settings smaller than Brazil’s $\sim 5,500$ municipalities should verify the small-sample behavior or implement a bias correction; for the Brazilian application this is not necessary.

6.5 Computational comparison with the full-enumeration formulation

The original full-enumeration formulation (Su, 2014) applied to the Brazilian setting would require enumerating $4^{5,570} \approx 10^{3,355}$ joint action profiles per parameter evaluation — an upper bound that is impossible by orders of magnitude. The mean-field formulation evaluates a J -dimensional fixed point in Δ^J at constant cost in I . Table 3 contrasts wall-clock estimation times across the two formulations on the same hardware.

Table 3. Computational cost: full-enumeration vs. mean-field formulation

	Full enumeration ($I = 3, J = 2, T = 1,000$)	Mean-field ($K = 30, J = 3, n = 5,570$)
Profiles enumerated	$4^3 = 64$	0 (no enumeration; fixed point only)
NLP variables	~ 50	~ 33 (or ~ 159 in MPEC form)
Single replication	0.05–1.9 s (SLSQP)	0.0002 s
$R = 500$ replications	7.4 s	1.4 s
Empirical-scale feasibility	infeasible (Brazil has $I = 5,570$)	feasible

Notes: Full-enumeration timings are from Section C and the Python re-implementation in `code/python/mc/`. Mean-field timings are from `code/python/mc.v2/`. Both formulations were run on the same laptop CPU, single-threaded, with the same Python and `numpy` versions. The full-enumeration formulation cannot be applied to Brazil’s $I = 5,570$ municipalities at any cost; the empirical-scale entry is shown for completeness.

The mean-field formulation is therefore not merely faster but qualitatively differently applicable: it is the only one of the two that can be brought to the Brazilian data.

6.6 Standard-error calibration

The 95% coverage rates of 84–91% in Table 1 reflect a small-sample limitation of single-stage HC0 standard errors: they treat the first-stage $\hat{\sigma}$ as known and have known downward bias when the number of moment cells is modest. We re-run the $R = 500$ replications under five alternative variance estimators — HC0, HC1, HC2, HC3 (MacKinnon and White, 1985), and a Pesendorfer–Schmidt–Dengler (Pesendorfer and Schmidt-Dengler, 2008) (PSD) structural-delta variance.³ Table 4 reports the 95% coverage for each.

³The PSD variance we use, $\widehat{\text{Var}}(y_\tau) = 1/[\hat{\sigma}(\tau, \hat{j}^*) \hat{\sigma}(\tau, 0) N_\tau]$, is the binary-outcome delta-method form generalized mechanically to $J > 1$ candidates by treating the winner-vs-not-winner contrast as a Bernoulli draw with probability $\hat{\sigma}(\tau, \hat{j}^*)$ and complement $\hat{\sigma}(\tau, 0)$. This is conservative relative to the full multinomial delta-method variance $[1/\hat{\sigma}(\tau, \hat{j}^*) + 1/\hat{\sigma}(\tau, 0)]/N_\tau$, which is the correct asymptotic when losing-candidate shares are non-trivial. Under our model the loser shares satisfy $\hat{\sigma}(\tau, \text{loser}) = \hat{\sigma}(\tau, 0)$ in expectation, so the binary form coincides with the multinomial limit; the slight over-coverage we observe in the Monte Carlo (§6.6) reflects this conservatism. We adopt the binary form throughout for transparency and computational simplicity.

Table 4. 95% coverage by SE family ($K = 30$, $n = 5,570$, $R = 500$)

Parameter	HC0	HC1	HC2	HC3	PSD
β_1	90.2%	91.4%	91.4%	93.8%	96.6%
β_2	90.6%	92.4%	93.0%	94.8%	97.0%
β_3	89.8%	92.6%	94.0%	95.8%	97.0%
$\alpha \cdot B_{j^*}$	84.6%	87.6%	87.8%	89.0%	93.6%
Mean SE / MC SD ratio (averaged across params)	0.86	0.92	0.94	1.00	1.06

Notes: HC1–HC3 are the small-sample-corrected variants of the heteroskedasticity-robust sandwich ([MacKinnon and White, 1985](#)): HC1 applies a $K/(K-r)$ degrees-of-freedom factor; HC2 divides each squared residual by $1-h_\tau$ (leverage); HC3 by $(1-h_\tau)^2$. PSD replaces the empirical residual by the structural delta-method variance derived from the binomial-margin first stage. The single best family for the structural parameter α is PSD; for β , HC3 is closest to nominal coverage. We adopt PSD as the default reported SE in the empirical exercise (Section 7) and report HC3 as a robustness check. Disagreement between PSD and HC3 in the empirical sample is informative: it can reflect either model misspecification — the empirical residual carries variance the structural form omits — or numerical sensitivity of the structural form when $\hat{\sigma}$ is close to zero or one for some cells.

The PSD column achieves 93.6–97.0% coverage — on or slightly above the nominal 95%. HC3 achieves 93.8–95.8% for β and 89.0% for α . The small over-coverage of PSD for β reflects a Laplace-smoothing inflation of $\hat{\sigma}(1-\hat{\sigma})$ for sparse cells; this inflation is conservative and does not bias the point estimates.

For the constrained-MLE estimator of [Su and Judd \(2012\)](#), the Hessian-based asymptotic SE is asymptotically efficient and exactly nominal at large n . For the two-step estimator we adopt as the headline approach, PSD with HC3 as a robustness check is our recommended default.

6.7 Summary

The mean-field formulation of the endorsement game admits a tractable, consistent estimator with $\sqrt{n}$ convergence at the sample sizes relevant to the Brazilian application. Bias is essentially zero at $n = 5,570$, RMSE is on the order of 0.05–0.07 for the structural parameters, and runtime is sub-millisecond per replication. The full-enumeration formulation that prior work has applied to a 3×2 toy problem cannot be extended to the Brazilian scale at any computational cost; the mean-field formulation is what makes the empirical application feasible.

7 Empirical application: Brazilian 2014 cycle

This section reports first-pass structural estimates of the v2 model on Brazilian 2014 data using a coalition-based weak proxy for mayoral endorsements. The estimates are correlational rather than causal: the proxy assigns each mayor to whichever presidential candidate’s coalition the mayor’s party formally joined, which mechanically captures formal alliance

behavior but understates the heterogeneity in actual public endorsement decisions that the model is designed to study. We treat these results as a feasibility benchmark for the framework and as a baseline against which the news-corpus-extracted endorsement variable of forthcoming work (Section 7.4) will be compared.

The headline structural parameter $\alpha \cdot B_{j^*}$ (the marginal utility of the budgetary prize, identified up to scale within a single cycle) is estimated in the range $[0.56, 0.65]$ across three bloc-augmented type partitions, with PSD t -statistics from 9.44 to 12.69. A bloc-free state-only baseline yields $\hat{\alpha} \cdot \hat{B} = 0.29$, which we read as omitted-variable bias once bloc identity is the dominant within-state predictor of endorsement (Section 7.2). An external anchor for B via the alignment-premium logic of Brollo et al. (2013) delivers a null result on parliamentary-amendment data, which we interpret as evidence that the coalition-based proxy is too noisy to identify the discretionary-transfer alignment effect that Brollo et al. (2013) document under a regression-discontinuity design.

7.1 Data construction

The estimation sample is constructed from three public sources joined at the mayor / municipality level.

TSE candidate registry (mayors). The 2012 mayoral election registry of the Tribunal Superior Eleitoral lists 5,569 municipalities with elected mayors who serve from 1 January 2013 through 31 December 2016 — the term in office during the 2014 presidential race. Of these, the merged sample retains 5,657 mayor-records after deduplicating by stable CPF identifier (a small fraction of mayors appear in both first-round and runoff records).

TSE coalition string (proxy endorsements). For each presidential candidate in 2014, the registry records the formal coalition string, i.e., the set of parties that joined that candidate for the first round. We construct a weak endorsement proxy by labelling mayor i as endorsing candidate j if mayor i ’s party either is candidate j ’s party (“strong” confidence, 35.7% of the sample) or is in candidate j ’s formal coalition (“weak” confidence, 64.3%). For 2014 every mayor’s party is in some coalition — formal coalitions in 2014 spanned the entire party landscape — so the proxy assigns every mayor to a candidate. The implied aggregate distribution: 60.0% to Dilma (PT, the actual winner), 24.8% to Aécio Neves (PSDB), 11.5% to Marina Silva (PSB), with the remainder to minor candidates.

IBGE municipality population. The 2014 `estimativa_dou` (Diário Oficial da União population estimate) gives municipality-level population at year-end 2014. We join the TSE and IBGE data via (state, normalized municipality name), achieving a 99.4% match rate (5,623 of 5,657 TSE munis matched). Unmatched cases reflect Portuguese-accent nor-

malization edge cases and a handful of post-2012 municipality boundary changes. Population is logged before standardization; population terciles are computed within each state.

Tesouro parliamentary-amendment disbursements. The Emendas Parlamentares dataset (Tesouro Transparente CKAN portal, 2015–2026 coverage) records every individual or bench-block parliamentary-amendment disbursement to a Brazilian municipality, including the IBGE code, year, value, and amendment type. We aggregate to (muni, year)-level totals and use post-election years 2015–2018 (Dilma’s second term) for the alignment-premium analysis in Section 7.3.

The full data-construction pipeline is documented in `docs/data_inventory.md` and is reproducible from raw TSE / IBGE / Tesouro files via the scripts in `code/python/data/`.

7.2 Type partitions and structural estimates

We estimate the v2 two-step OLS estimator under four progressively richer type partitions, all with the binary endorsement outcome $j = 1$ (endorses Dilma) versus $j = 0$ (endorses anyone else, including non-endorsement). Type-conditional log-odds are regressed on type-level covariates and a constant, with the constant identifying $\alpha \cdot B_{j^*}$ under the normalization $B_{j^*} = 1$. Standard errors are reported as Pesendorfer–Schmidt–Dengler (PSD) structural-delta variance per Section 6.6.

The four specifications differ in the granularity and content of the type partition:

- **(i) State only ($K = 26$):** types are the 26 Brazilian states (DF excluded — it lacks a mayoral office in the standard sense). Type covariates $X(\tau)$ are state-level mayor demographics: mean age, fraction female, fraction with higher education. The simplest specification, but state demographics are weakly predictive of mayoral endorsement.
- **(ii) State $\times$ 3-bloc ($K = 75$):** mayors are additionally partitioned by ideological party bloc using the standard Power–Zucco classification (Left / Center / Right). Two bloc dummies enter $X(\tau)$ with Center as reference.
- **(iii) State $\times$ 4-bloc ($K = 96$):** the Left bloc is split into “PT-aligned” (PT, PDT, PCdoB, PPL, PSOL) and “Marina-aligned” (PSB, PV) sub-blocs. The split is motivated by within-Left endorsement heterogeneity in 2014: PT-aligned mayors endorsed Dilma at 98.7%, Marina-aligned mayors at 0%. With three bloc dummies and Center as reference, this specification dramatically improves fit.
- **(iv) State $\times$ 4-bloc $\times$ pop-tercile ($K = 222$):** types are further partitioned by within-state population tercile, and $\log(\text{population}_\tau)$ is added to $X(\tau)$. Population weights $w(\tau)$ are set to type’s share of the national population, satisfying the v2 model’s population-weighted aggregation in Section 2.1.

Table 5 reports the four specifications side by side.

Table 5. Two-step OLS estimates across four type partitions, 2014 cycle

Specification	(i) State only	(ii) State $\times$ 3-bloc	(iii) State $\times$ 4-bloc	(iv) State $\times$ 4-bloc $\times$ pop-tercile
<i>Structural budget-prize parameter</i>				
$\alpha \cdot B_{j^*}$	0.274 (0.028)	0.584 (0.050)	0.574 (0.062)	0.647 (0.051)
t -statistic	9.90	11.65	9.28	12.69
<i>Bloc fixed effects (Center is reference)</i>				
$\beta_{\text{bloc, Left}}$	—	−0.48 (0.128)	—	—
$\beta_{\text{bloc, Left-PT}}$	—	—	2.56 (0.302)	2.09 (0.219)
$\beta_{\text{bloc, Left-PSB}}$	—	—	−4.63 (0.353)	−4.17 (0.262)
$\beta_{\text{bloc, Right}}$	—	−0.99 (0.116)	−0.98 (0.120)	−1.13 (0.113)
<i>Demographic and economic controls</i>				
$\beta_{\text{mean age}}$	0.10 (0.124)	0.04 (0.064)	0.10 (0.119)	0.03 (0.080)
$\beta_{\text{frac female}}$	−0.10 (0.117)	−0.11 [†] (0.066)	−0.02 (0.126)	−0.00 (0.082)
$\beta_{\text{frac higher ed}}$	−0.10 (0.122)	0.04 (0.071)	−0.08 (0.144)	−0.17 [†] (0.094)
$\beta_{\log(\text{pop})}$	—	—	—	−0.02 (0.083)
Number of types K	26	75	95	222
R^2	0.086	0.181	0.887	0.840
Number of mayors n	5,623	5,616	5,604	5,443

Notes: Outcome is $\log(\hat{\sigma}(\tau, \text{Dilma})/\hat{\sigma}(\tau, \text{not-Dilma}))$. Estimator is two-step OLS on type-conditional Laplace-smoothed empirical choice probabilities (see Section 5.1). All four columns use the same TSE-IBGE-joined sample frame and apply population weights $w(\tau) = \text{pop_share}(\tau)/f(\tau)$ so that population-weighted aggregate shares satisfy $\sum_{\tau} f(\tau)w(\tau) = 1$; numbers are accordingly directly comparable across columns. Sample sizes vary slightly across columns because some cells fail the ≥ 5 mayors-per-cell filter at finer partitions (specifications iii-iv) and because mayors with parties absent from the Power-Zucco bloc lookup (54 PCdoB and 25 PTdoB total in 2014) are dropped only from bloc-augmented specifications. Standard errors in parentheses are PSD structural-delta variance (Section 6.6) throughout. The [†] marker indicates marginal significance ($0.05 < p < 0.10$); all reported $\hat{\alpha} \cdot B$ and bloc coefficients without a marker are significant at $p < 0.001$. $K = 26$ is the number of states excluding the Federal District (no mayoral office). Center (PMDB / PSDB / PSD / PPS / PROS / REDE) is the reference bloc throughout. Reproduces from `code/python/data/estimate_2014_harmonized.py`.

The point estimate of $\hat{\alpha} \cdot B_{j^*}$ is 0.274 in the state-only specification (i) and rises to 0.584, 0.574, and 0.647 in specifications (ii) through (iv). The state-only estimate is biased downward by an omitted-variable problem: bloc identity is the dominant predictor of mayoral endorsement, so a specification that does not partition by bloc forces the structural-prize term to absorb between-bloc variation that should belong to the bloc fixed effects. Once bloc dummies are added in specifications (ii)–(iv), $\hat{\alpha} \cdot B$ stabilizes in the range $[0.56, 0.65]$ across an order-of-magnitude variation in K . We treat 0.56–0.65 as the structural-parameter range identified by the §1.1 proxy data; the spec-(i) estimate is reported for transparency. The t -statistic exceeds 9 in all four specifications under the PSD structural-delta variance, reflecting tight identification once any bloc partition is in place.

The bloc fixed effects exhibit the expected coalition pattern. PT-aligned mayors are roughly $\exp(2.09) \approx 8\times$ more likely (in odds) to endorse Dilma than the Center reference; Marina-aligned (PSB / PV) mayors are roughly $\exp(-4.17) \approx 0.015\times$ as likely; and Right-bloc mayors are roughly $\exp(-1.13) \approx 0.32\times$ as likely. These are not behavioural surprises — they trace the formal coalition assignments of the proxy directly — but they confirm that the model’s partition-by-bloc identification works at the expected magnitudes.

The demographic controls are uniformly small. After controlling for state and bloc, mean age, fraction female, and log population are not statistically significant. The fraction-of-mayors-with-higher-education coefficient turns marginally significant in the most granular specification ($\hat{\beta} = -0.17$, $p = 0.07$), suggesting that within state $\times$ bloc $\times$ population tercile cells, more-educated mayors lean somewhat against the PT incumbent — consistent with the urban PSDB-aligned character of high-education municipalities. These secondary patterns are sensitive to specification choice and we treat them as suggestive rather than definitive.

The high R^2 values in specifications (iii) and (iv) (0.84–0.89) are partly mechanical: the coalition-based proxy is itself a deterministic function of the mayor’s party, so once the type partition encodes party bloc, fit is essentially perfect within Type-I-extreme-value sampling noise. The much lower $R^2 = 0.18$ of specification (ii) is informative — it tells us that aggregating Left into a single bloc (PT and PSB pooled) loses most of the explanatory power, because PT and PSB mayors endorse opposing candidates in 2014.

Robustness: leave-one-bloc-out. Because the bloc partition does most of the explanatory work in specifications (iii) and (iv), a natural concern is that $\hat{\alpha} \cdot B_{j^*}$ is identified almost entirely from the mechanical bloc structure of the proxy rather than from genuine within-bloc variation. To test this, we re-estimate specification (iv) on four leave-one-out subsamples that drop, in turn, the four mayoral blocs. Table 6 reports the result.

Table 6. Leave-one-bloc-out robustness for specification (iv)

Sample	Mayors	Cells K	$\hat{\alpha} \cdot B_{j^*}$	PSD SE	t	R^2
Full sample (spec iv)	5,623	222	0.647	0.051	12.69	0.840
Drop PT-aligned mayors	4,604	171	0.522	0.039	13.28	0.770
Drop PSB-aligned mayors	5,076	184	0.721	0.056	12.84	0.636
Drop Center mayors [†]	3,263	154	−0.032	∞	0.00	0.898
Drop Right-bloc mayors	3,926	157	0.670	0.057	11.73	0.885

Notes: PSD structural-delta variance per Section 6.6. The Center bloc is the reference category in specification (iv), so dropping it makes the bloc-dummy regression rank-degenerate and $\hat{\alpha}$ algebraically unidentified ([†]); we report this case for completeness only. Across the three non-degenerate subsamples, $\hat{\alpha} \cdot B_{j^*}$ ranges from 0.522 (drop PT-aligned, −19% of full) to 0.721 (drop PSB-aligned, +12% of full), with the full estimate 0.647 near the middle of the range. Reproduces from `code/python/data/estimate_2014_robustness.py`.

The estimate of $\hat{\alpha} \cdot B_{j^*}$ moves by at most 19% across the three non-degenerate leave-one-out subsamples, with the full estimate sitting comfortably inside this range. The structural parameter is therefore not driven by any single bloc’s mechanical contribution to the proxy; rather, the within-bloc variation in $\hat{\sigma}(\text{Dilma} \mid \tau)$ between state-and-population-tercile cells of the same bloc supplies the residual signal that pins down the constant in the log-odds regression. We do not, however, claim that this rules out specification dependence at the broader level. The fact that $\hat{\alpha}$ rises when PSB-aligned mayors (uniformly 0% Dilma in the proxy) are dropped, and falls when PT-aligned mayors (uniformly 99% Dilma in the proxy) are dropped, suggests the estimate continues to weight extreme-bloc observations more heavily than equal-influence regression theory would prefer. The richer proxy variation that a news-corpus extraction (Section 7.4) would provide should reduce this sensitivity.

7.3 An external anchor for B : a null result

Within a single cycle the structural model identifies only the product $\alpha \cdot B_{j^*}$, not α and B separately. The classical strategy for separately anchoring B uses post-election fiscal alignment: the budgetary prize a winning candidate delivers to her aligned mayors should be measurable in realised intergovernmental transfers. We attempt this following [Brollo et al. \(2013\)](#) and using the parliamentary-amendment data.

For each Brazilian municipality we compute total per-capita parliamentary-amendment receipts during 2015–2018 (Dilma’s second term) and regress the log on a Dilma-coalition indicator with population, FPM-per-capita, and state fixed-effects controls. The sample is the same 5,623-muni TSE-IBGE-joined frame used in Section 7.2; the additional Emendas and FPM merges drop only one muni in total (FPM unmatched under the spelling-normalised merge key), so the regression sample is 5,622. Table 7 reports nine specifications spanning two robustness dimensions: temporal aggregation (full term vs. 2015 only) and political clarity (full sample vs. PT-vs-Aécio mayors only).

Table 7. Alignment-premium regressions for the 2014 cycle

Spec	Outcome window	Sample	Controls	$\hat{\gamma}_{\text{align}}$	t	R^2
(1)	2015-2018	full	none	0.047 [†]	1.93	0.001
(2)	2015-2018	full	+ log(pop)	0.026	1.40	0.423
(3)	2015-2018	full	+ log(FPM/cap)	0.022	1.19	0.428
(4)	2015-2018	full	+ state FE	0.006	0.33	0.531
(5)	2015-2018	full	+ state FE + log(FPM/cap)	0.007	0.38	0.531
(6)	2015 only	full	+ log(pop)	0.094 [*]	2.09	0.026
(7)	2015 only	full	+ log(pop) + state FE	0.051	1.12	0.110
(8)	2015-2018	PT-vs-Aécio	+ log(pop) + state FE	0.002	0.08	0.537
(9)	2015-2018	PT-vs-Aécio	+ log(pop) + state FE + log(FPM/cap)	0.003	0.12	0.537

Notes: Outcome is $\log(1 + \text{per-capita parliamentary amendments BRL})$. “Aligned” indicator equals one when the mayor’s party is in Dilma’s coalition. Sample sizes: full = 5,622 munis; PT-vs-Aécio (mayors whose proxy endorses Dilma or Aécio Neves only) = 4,763. HC0 (heteroskedasticity-robust) standard errors. We do not cluster standard errors at the state level, although mayoral earmark outcomes within a state are not independent (a state’s federal-deputy delegation operates as a single political-economy unit). State-clustered errors would widen the alignment-premium intervals further and push the central estimates toward zero — in the same direction as our reported null. We adopt unclustered HC0 here for transparency with respect to the within-cell sampling variability and report the broader inference issue as a known limitation. [†] $p < 0.10$; ^{*} $p < 0.05$.

The full-controls specification (5) yields $\hat{\gamma}_{\text{align}} = 0.007$ ($p = 0.71$), economically and statistically indistinguishable from zero. The most permissive specification (1) yields 0.047 ($p = 0.054$), an upper bound. The 2015-only specification (6) is the most informative single estimate: $\hat{\gamma}_{\text{align}} = 0.094$ ($p < 0.05$) without state fixed effects, dropping to 0.051 once state fixed effects are included. We read this 9.4% point estimate as a lower-bound on the true alignment premium under our coarse proxy: it captures the share of mayors whose true alignment was both (i) sharp enough to register on a coalition-based proxy and (ii) stable enough to translate into 2015 earmark receipts before the post-2016 institutional realignment. The clean PT-vs-Aécio comparison (specifications 8–9) returns null estimates indistinguishable from zero.

These point estimates fall far below the 30–40% premium that [Brollo et al. \(2013\)](#) identify under their close-mayoral-elections RDD design. The numerical comparison is suggestive but not strict: [Brollo et al. \(2013\)](#) estimate a local average treatment effect at the 50-50 close-margin cutoff, whereas our $\hat{\gamma}_{\text{align}}$ is an OLS estimand on the full population of mayors with non-trivial selection into proxy-aligned status. The two estimands differ by both estimator (LATE vs OLS) and effective sample (margin-of-victory cutoff vs full sample), so the gap is not exclusively a measure of proxy-quality attenuation. We diagnose the broader pattern of small alignment-premium estimates across our specifications as driven by three factors:

1. **The proxy is too coarse.** PMDB and other Centrão mayors are mechanically labelled as “aligned” with Dilma in 2014 (because PMDB joined Dilma’s coalition with

Temer as VP), yet their actual political behavior — and their access to discretionary federal money — is famously coalition-independent. With $\sim 35\%$ of the proxy assignments this noisy, the alignment indicator is a weak instrument for the underlying object [Brollo et al. \(2013\)](#) measure.

2. **State fixed effects absorb most of the variation.** The drop in $\hat{\gamma}_{\text{align}}$ from 0.047 in specification (1) to 0.007 in specification (5) reflects regional confounding: north-eastern and northern states are simultaneously more PT-aligned (proxy) and more dependent on federal earmarks (size of federal-deputy delegations), so the cross-sectional alignment correlation is largely geographic.
3. **The post-2016 impeachment regime change degrades the signal.** PMDB officially broke with the PT government in March 2016, after Dilma’s 2014 coalition was already obsolete. Our 2015-only specification (6) shows a stronger premium (9.4%, $p < 0.05$) consistent with this story; pooling 2015–2018 dilutes the 2015 signal with three years of post-impeachment realignment that the proxy mechanically fails to capture.

We accordingly do *not* adopt this $\hat{\gamma}_{\text{align}}$ as an anchored estimate of B_{2014} in the structural model. The exercise is informative as a falsification check on the proxy: it tells us that the headline $\hat{\alpha} \cdot B_{j^*} \in [0.56, 0.65]$ point estimate of Section 7.2 is identified by within-bloc variation in $\hat{\sigma}$ at finer type levels rather than by an externally calibrated B , and that anchoring B from outside the model awaits cleaner endorsement data. In a multi-cycle exercise (Section 7.4), α and B separate via cross-cycle structural variation in candidate identities (paper Section 3.3); the panel exercise is the natural next step.

7.4 Limitations and the case for the news-corpus extension

The single-cycle Brazilian exercise has three limitations the v2 framework is designed to address but that this first-pass implementation does not. We register them here as the Brazilian-side follow-up agenda; Section 8 pursues a parallel French scope-condition exercise that addresses a separate question — whether the framework’s prize-driven mechanism applies in institutional settings where the endorsement signal is individual rather than party-mediated.

Single-cycle data. With one cycle, α and B are identified only as a product. A two-cycle panel (2014 and 2018, or 2014 and 2022) suffices to identify them separately given variation in candidate composition. Adding 2018 and 2022 is straightforward as a code extension (see paper Section 3.3 for the panel identification argument and Section 6.1 for the Monte-Carlo benchmark of cross-cycle precision). The relevant empirical question is whether the 2018 and 2022 weak proxies are sharp enough for the cross-cycle exercise. Our descriptive look at

the proxy in those cycles (`code/python/data/build_proxy_endorsement.py` output, see `docs/data_inventory.md`) suggests not: the 2018 proxy assigns Geraldo Alckmin 52% of mayoral endorsements via the formal Centrão coalition, when his actual first-round vote share was 4.8%. The proxy substantially overstates Centrão-coalition discipline.

Mechanical proxy structure. Our $R^2 = 0.84\text{--}0.89$ values in specifications (iii)–(iv) are inflated relative to the actual variability in mayoral endorsement decisions, because the proxy *is* the bloc structure under a different label. Real-world endorsements involve substantial within-bloc cross-cutting — Centrão mayors splitting between candidates, ideologically-aligned mayors withholding endorsement, and PT mayors occasionally endorsing PSB or PCdoB candidates. The structural model’s potential insight — recovering $\hat{\alpha}$ from the residual variation after bloc averages — is hidden when the proxy is too tight.

Further geographic robustness. State fixed effects in Table 5 absorb only state-level confounding. Brazilian mayoral coalition behaviour varies meaningfully *within* states by sub-state region (semi-arid Northeast vs. coastal Northeast, for instance), and the IBGE meso-region (*mesorregião*) and micro-region (*microrregião*) classifications provide the natural finer geographic strata. The data inventory in `docs/data_inventory.md` §4 documents these as available IBGE variables; we did not include them in the present specifications because they cross-cut the state-bloc partition and would interact non-trivially with the population-tercile cells. Replicating Table 5 with meso-region fixed effects in place of state fixed effects, or with both, is a natural robustness exercise to add in the empirical extension once the news-corpus endorsement variable replaces the proxy.

The news-corpus pipeline. Both limitations dissolve under news-extracted endorsement data. The plan, documented in `docs/data_sources.md` and `docs/roadmap.md` (Phase 3 weeks 4–8), is:

1. Compile a news corpus from the major Brazilian outlets (Folha de São Paulo, O Globo, Estadão, Poder360) covering the six-month pre-election windows in 2014, 2018, and 2022, supplemented by GDELT 2.0 and (for 2014–2018) historical Twitter archives.
2. Train a Portuguese named-entity-recognition pipeline (spaCy `pt_core_news_lg` or BERTimbau fine-tuned for relation extraction; alternatively, an LLM few-shot extractor) to extract (`mayor`, `candidate`, `support / opposition / appearance`, `date`) triples from text.
3. Validate against a hand-coded ground-truth subset from the top 30 Brazilian municipalities, where endorsement records are well documented from secondary sources.

4. Re-run the structural estimation with the news-extracted endorsement variable replacing the proxy.

The expected gain is twofold. First, the cross-bloc heterogeneity that the proxy cannot capture — centrão defection, ideologically-aligned non-endorsement, and intra-Left endorsement realignment — becomes observable, and the structural model’s $\hat{\alpha}$ becomes a measurement of behavioural responsiveness rather than coalition-discipline mechanics. Second, the news-extracted endorsement variable is sufficiently sharp at the muni level that the alignment-premium regression of Section 7.3 should recover a non-null $\hat{\gamma}_{\text{align}}$ closer to Brollo et al. (2013)’s magnitudes, anchoring $B_{j,c}$ from outside the model and providing a triangulation of the structural estimate. Both extensions are tractable conditional on the news corpus being assembled.

8 Scope condition: French parrainages, 2017 and 2022

This section applies the structural framework to a second institutional setting — the French parrainages system — and uses the contrast with the Brazilian application of Section 7 to delineate the institutional preconditions under which the prize-driven mean-field mechanism captures observed endorsement behavior. The French setting is empirically attractive for three reasons: (i) every authorized elected official’s parrainage is signed personally and published verbatim by the Conseil constitutionnel, eliminating measurement error in the endorsement variable; (ii) two presidential cycles (2017 and 2022) supply panel variation; and (iii) the same incumbent (Macron) wins both, so the structural object “alignment with the eventual winner” is invariant across cycles and within-mayor switching across cycles is informative. Yet the French data reveal that the prize-driven mean-field model does *not* fit French parrainages, a finding we trace to a clean institutional difference from Brazil: in France, the endorsement signal is individual, ideological, and tactically tied to the 500-signature ballot-qualification threshold rather than party-mediated and prize-maximizing.

The headline numerical contrast: the alternative-specific extension of the v2 estimator (Section 5.3) recovers ideologically-interpretable per-candidate covariate effects on the French data — Macron’s mayor base is urban + peri-urban, Mélenchon’s is dense urban, Pécresse’s is rural lower-population, all matching documented voter-base patterns — but the structural budget-prize parameter $\hat{\alpha} \cdot \hat{B}$ collapses mechanically to -0.121 in 2017 and -0.127 in 2022, with PSD t -statistics around -50 in both cycles. The mechanical negative arises from the equilibrium share equation: with Macron’s mayor-sponsorship share $\hat{s}_{\text{Macron}} \approx 0.04$, the model implies the average log-odds $\overline{\log[\hat{\sigma}(\tau, \text{Macron})/\hat{\sigma}(\tau, 0)]} \approx -2.9$ must be reconciled by $\alpha \cdot B/s_{\text{Macron}}$, yielding $\alpha \cdot B \approx -0.12$ irrespective of the per-candidate covariate effects. Macron’s small mayor share is itself the data’s response to the model’s question: *if the prize were positive and material, more mayors would pile onto endorsing*

the winner; that pile-on does not occur in French parrainages. We interpret this as a sharp scope condition rather than a model failure — the framework correctly identifies that French parrainage behavior is incompatible with a prize-dominant payoff structure.

8.1 Data construction

The French sample is constructed from three public sources joined at the commune level.

Conseil constitutionnel parrainages. The Conseil constitutionnel publishes the validated parrainages for each presidential cycle as open-data CSV on data.gouv.fr. We pull both the 2017 and 2022 cycles (14,296 and 13,427 total parrainages respectively, of which 10,310 and 9,207 are from elected mayors — the rest are département / regional councilors and senators / deputies). Each parrainage record gives the signer’s civility, name, role (*mandat*), commune (*circonscription*), département, candidate sponsored, and date of validation. The schema is identical across cycles up to a column-name change for the candidate field; we harmonize and pool. The endorsement variable is sharp: a mayor either signed a parrainage for exactly one candidate, or did not sign — there is no measurement-error layer of the kind that [Ferraz and Finan \(2008\)](#) discuss for media-based endorsement extraction.

INSEE commune master. Each parrainage identifies the commune by free-text name only; to attach population, geography, and urban-density covariates we resolve commune names to the INSEE 5-digit *code officiel géographique* (COG) using the data.gouv.fr-curated commune master file (34,935 commune records, 47 columns, including four orthographic-variant name forms for fuzzy matching). The match strategy restricts INSEE candidates to the parrainage’s département, then attempts exact match on each of the four name variants in turn, after a normalization that strips accents, hyphens, leading articles, and case. Match rate on mayor parrainages is 97.2% (18,977 of 19,517 matched); the 540 unmatched are concentrated in overseas territories (Polynésie / Nouvelle-Calédonie, 213 rows; partially absent from the metropolitan COG file) and Paris arrondissement naming conventions. We drop unmatched mayors from estimation.

Universe definition. The structural model requires a denominator: the universe of mayors who *could have* signed a parrainage in cycle c . We use the INSEE commune master itself (filtered to ordinary communes COM and Paris/Lyon/Marseille arrondissements ARM), giving a per-cycle mayor universe of 34,926. Each commune’s mayor in cycle c is then classified as *sponsoring Macron* (the cycle’s eventual winner), *sponsoring some other qualifying candidate* (one who received ≥ 500 total parrainages and so was on the ballot), or *abstaining* (no parrainage signed, plus the 1,000–1,200 per cycle who sponsored a non-qualifying fringe candidate). For 2017 we have 1,352 Macron sponsors, 7,101 other-qualifying sponsors across

$J - 1 = 10$ losing qualifiers, and 26,473 abstainers (including 1,234 fringe-sponsor mayors folded into abstention). For 2022 we have 1,193 Macron sponsors, 6,578 other-qualifying sponsors across $J - 1 = 11$ losing qualifiers, and 27,155 abstainers (including 1,040 fringe-sponsor mayors). The population-weighted aggregate Macron mayor-sponsorship share that enters the structural estimator is $\hat{s}_{\text{Macron}} = 0.041$ in 2017 and 0.040 in 2022.

DGCL discretionary investment grants. The DGCL publishes annual project-level files for every state grant disbursed to a commune under one of the four discretionary investment-support schemes: DETR (Dotation d’Équipement des Territoires Ruraux), DSIL (Dotation de Soutien à l’Investissement Local), DPV (Dotation Politique de la Ville), and DSID (Dotation de Soutien à l’Investissement des Départements). These are not formula-based; they are awarded by the prefect, who is the president’s regional representative, on a project-by-project basis, which makes them the natural French analog of the discretionary-transfer channel that [Brollo et al. \(2013\)](#) document for Brazilian emendas parlamentares. We pull seven annual files (2018–2024, $\approx 26,000$ project rows per year) and aggregate to (commune, year, scheme) totals. Total disbursement across the seven years is approximately EUR 10.4 billion (DETR EUR 5.46 B + DSIL EUR 4.02 B + DPV EUR 0.85 B + DSID EUR 0.07 B); 90% of 2017-cycle Macron-aligned communes received some grant in their post-cycle window.

The full data-construction pipeline is reproducible from the four scripts named in the file header. Reduced-form alignment-premium estimates appear in the long working version; we summarize in [Section 8.2](#).

8.2 Reduced-form alignment premium

Before the structural exercise, we compute the reduced-form alignment premium following the Brollo logic: regress per-capita post-cycle DETR + DSIL + DPV + DSID receipts on a Macron-aligned dummy with controls for log population, log density, and département fixed effects. The pooled-cycle estimate (cycle FE plus département FE) is $\hat{\gamma}_{\text{align}} = +0.105$ (HC1 SE 0.037, $p < 0.01$), an $\exp(0.105) - 1 \approx 11\%$ premium in per-capita grants for Macron-aligned mayors. The 2022-cycle estimate alone is $\hat{\gamma} = +0.176$ ($p < 0.01$, $\approx 19\%$ exponentiated); the 2017-cycle estimate alone is $\hat{\gamma} = +0.041$ ($p = 0.39$, not significant). The cycle pattern is consistent with Macron’s LREM being a new party in 2017 with weak prefectural networks, with prefectural alignment consolidating by 2022 as the Macroniste apparatus matured. A within-mayor difference using the 3,115 mayors who signed in both cycles — of whom 208 switched alignment status — yields a null point estimate ($\hat{\gamma} = +0.015$, not significant). The within-mayor null is informative: it suggests the cross-sectional alignment premium is partly driven by selection on persistent commune characteristics that predict both Macron-sponsorship and grant flow, which is precisely the selection problem

the structural model is designed to undo.

The reduced-form alignment premium is positive and material at the population-cross-section level. If the v2 mean-field model fits France, the structural recovery of $\alpha \cdot B$ should be *positive* and approximately commensurate with what mayors expect from a $\sim 11\%$ multiplicative premium on annual discretionary grants. The next subsection shows that no such recovery occurs.

8.3 Structural estimates: alternative-specific extension

We estimate the alternative-specific extension of the two-step OLS estimator (Section 5.3). The estimator allows each candidate j to have its own coefficient vector β_j on the type-level covariate vector $X(\tau)$:

$$\log \left[\frac{\hat{\sigma}(\tau, j)}{\hat{\sigma}(\tau, 0)} \right] = X(\tau)' \beta_j + \mathbf{1}\{j = \text{Macron}\} \frac{\alpha \cdot B}{\hat{s}_{\text{Macron}}} + \varepsilon_{\tau, j}, \quad j = 1, \dots, J, \quad (28)$$

estimated equation-by-equation by OLS. The winner equation (Macron) carries the additional $1/\hat{s}_{\text{Macron}}$ regressor that identifies $\alpha \cdot B$ jointly with β_{Macron} ; the $J - 1$ loser equations identify β_j for each loser separately. Type partition is the French département (`dep_code`, $K = 97$ in both cycles). Type-level covariates are mean log population, mean log density, and the share of communes in the type that are in an INSEE-classified urban unit of size $\geq 10,000$ (the `taille.unite_urbaine` ≥ 3 threshold), each standardized to mean 0 and standard deviation 1 before estimation. Population weights $w(\tau) = \text{pop_share}(\tau)/\hat{f}(\tau)$ as in Section 7. Standard errors are PSD structural-delta variance.

Table 8 reports the headline structural-prize result.

Table 8. Structural budget-prize parameter, French parrainages

Cycle	K	J	$\hat{s}_{\text{Macron}}$	$\hat{\alpha} \cdot \hat{B}$	PSD SE	PSD t
2017	97	11	0.041	-0.121	0.002	-56.84
2022	97	12	0.040	-0.127	0.002	-51.78

Notes: Estimates from the alternative-specific extension of the two-step OLS estimator (equation 28), one row per cycle. $K = 97$ types are French départements; J is the number of qualifying candidates per cycle (those who received ≥ 500 total parrainages and so were on the ballot). Population weights $w(\tau) = \text{pop_share}(\tau)/\hat{f}(\tau)$. Standard errors are PSD structural-delta variance. The point estimates are mechanically negative; see Section 8.4 for the diagnosis. Reproduces from `code/python/data/estimate_france_v2.py`.

The point estimate of $\hat{\alpha} \cdot \hat{B}$ is sharply negative in both cycles. Mechanically, the model's identification equation for the winner reads $\overline{\log[\hat{\sigma}(\tau, \text{Macron})/\hat{\sigma}(\tau, 0)]} = \overline{X(\tau)' \beta_{\text{Macron}}} + (\alpha \cdot B)/\hat{s}_{\text{Macron}}$. With X standardized to mean 0 across types, the second term equals the

average log-odds. Empirically that average is $\overline{\log[0.04/0.71]} \approx -2.85$. With $\hat{s}_{\text{Macron}} = 0.041$, the implied $\alpha \cdot B = -2.85 \times 0.041 \approx -0.117$, very close to the table’s -0.121 . The negative sign is therefore not an estimator artifact but the model’s structurally-correct response to a population in which Macron-sponsorship is too rare to be reconciled with a positive structural prize: if the prize were positive, the equilibrium share equation would predict more pile-on than the data show.

The per-candidate β_j estimates, by contrast, are entirely sensible. Table 9 reports the alternative-specific covariate coefficients for the 2022 cycle (the 2017 cycle pattern is similar; full table available on request).

Table 9. Per-candidate covariate effects β_j , French 2022 parrainages

Candidate	$\hat{\beta}_{\log \text{ pop}}$	$\hat{\beta}_{\log \text{ dens}}$	$\hat{\beta}_{\text{urban share}}$	$\hat{s}_j$
Macron (winner)	+0.378 (0.219)	−0.440 (0.189)	+0.443 (0.244)	0.040
Pécresse (LR)	−0.455 (0.164)	+0.517 (0.244)	+0.763 (0.219)	0.084
Hidalgo (PS)	+0.169 (0.263)	−0.328 (0.276)	+0.728 (0.237)	0.029
Mélenchon (LFI)	−0.245 (0.246)	−0.422 (0.242)	+0.974 (0.290)	0.022
Le Pen (RN)	−0.561 (0.420)	+0.020 (0.416)	+0.789 (0.487)	0.009
Zemmour (Reconquête)	−0.200 (0.357)	−0.125 (0.339)	+0.551 (0.437)	0.020

Notes: Six of $J = 12$ qualifying candidates shown; the six minor-share candidates (Arthaud, Dupont-Aignan, Jadot, Lassalle, Poutou, Roussel) follow similar patterns and are omitted for brevity. Coefficients are on standardized covariates (mean 0, sd 1). PSD structural-delta SE in parentheses. $\hat{s}_j$ is the population-weighted aggregate mayor-sponsorship share. Reproduces from `code/python/data/estimate_france.v2.py`.

The covariate signature of each candidate matches the documented voter-base pattern. Macron’s mayor base concentrates in higher-population, lower-density, urban-share-positive départements — the peri-urban Macron coalition. Pécresse (LR) attracts smaller-population but higher-density départements — the LR rural-but-urban-dense pattern of the traditional French right. Hidalgo (PS) and Mélenchon (LFI) both load heavily on urban-share, with Mélenchon additionally favoring lower-density départements — the dense urban left versus the broader urban left split. Le Pen and Zemmour both load negatively on log population — consistent with the small-municipality far-right mayor base. None of the per-candidate effects is statistically distinguishable at the single-département level, but the pattern across the table closely matches documented département-level vote-share patterns from the same cycles (INSEE results).

The contrast between Tables 8 and 9 is the headline finding of this section. The alternative-specific estimator successfully *recovers* the ideological-matching component of the parrainage decision (Table 9); it simultaneously *rejects* the prize-driven component (Table 8). Both findings are produced by the same OLS regressions on the same data — they are not separable diagnostic exercises but two outputs of one estimation. The conclusion is that the v2 mean-field model’s prize-driven mechanism is mis-specified for French parrainages, while the alternative-specific extension recovers exactly the ideological-matching component that French parrainage behavior actually exhibits.

8.4 Diagnosis: which model assumption fails?

The mechanical algebra of $\hat{\alpha} \cdot \hat{B} \approx -0.12$ traces back to one feature of the data and one feature of the model that, in combination, are incompatible.

Data feature. Macron, the eventual electoral winner in both 2017 and 2022, is *not* the most-endorsed candidate in either cycle’s mayor universe. In 2017, Fillon (LR) received 2,013 mayor parrainages versus Macron’s 1,352; in 2022, Pécresse (LR) received 1,332 versus Macron’s 1,193. Macron’s aggregate mayor-sponsorship share $\hat{s}_{\text{Macron}} \approx 0.04$ is far below the most-endorsed candidate’s share of 0.10 (Fillon 2017) or 0.084 (Pécresse 2022). This is not noise — it is a stable institutional feature of French presidential mayor-sponsorship: traditional party-machine candidates (LR / formerly UMP) collect more mayor parrainages because mayors of smaller, more rural communes tend to belong to the LR party network, while Macron’s LREM / Renaissance is younger and weaker at the municipal level.

Model feature. The v2 mean-field equilibrium equation $\sigma(\tau, j) = \exp(\pi(\tau, j; s)) / \sum_k \exp(\pi(\tau, k; s))$ with $\pi(\tau, \text{winner}) = X(\tau)' \beta + \alpha \cdot B / s_{\text{winner}}$ predicts that, holding $\alpha \cdot B > 0$, s_{winner} should rise under iteration until the marginal mayor is indifferent. The model’s logical content is: *if the prize is positive, more mayors should pile onto endorsing the eventual winner, and this pile-on should be visible in the equilibrium share.* With $\hat{s}_{\text{Macron}} \approx 0.04$, the data offer no such pile-on. The model’s only consistent response is to recover $\alpha \cdot B \leq 0$, and that is exactly what Table 8 reports.

Which assumption fails. The candidate diagnoses are: (a) the prize is small or zero in France — mayors gain little from being on the winning team, so they do not pile on; (b) mayors do not perceive a meaningful prize from the parrainage signal even if one exists post-election — the parrainage is not the sort of public commitment that the post-election prefectural apparatus uses to allocate DETR/DSIL projects; (c) the parrainage decision is driven by a different motive entirely — ideological match plus the tactical 500-signature

ballot-qualification game — so the prize term is dominated by other motives and effectively zero in equilibrium.

The reduced-form alignment premium of Section 8.2 (+11% pooled, in per-capita grants for Macron-aligned mayors) rules out (a): there is a prize. Diagnoses (b) and (c) are both consistent with the data. Diagnosis (b) is partly ruled out by the cross-cycle pattern: the 2022 alignment premium is +19% (significant), suggesting that the prefectural allocation does respond to parrainage history once the Macroniste apparatus matures. Diagnosis (c) is the residual: the parrainage system is structured around a 500-signature qualification threshold, which gives mayors of small parties a tactical reason to sign for their party’s candidate *even when that candidate has no chance of winning*. Sponsoring Cheminade (3rd-place finisher with 1.21% of vote in 2017) or Lassalle (4.7%) is not a bid for the prize — it is a coordination decision about which candidates appear on the ballot. The model has no representation of this qualification motive; consequently the model’s prize-share equation cannot rationalize the observed equilibrium and the structural prize is rejected.

8.5 Discussion: scope conditions

The contrast between Brazil and France yields a sharper claim about the model’s domain of applicability than either application alone could deliver.

Where the model applies. The Brazilian application of Section 7 demonstrates that the v2 mean-field framework recovers a positive, precisely estimated structural prize ($\hat{\alpha} \cdot \hat{B} \in [0.56, 0.65]$, $t > 9$) when (i) the endorsement signal is mediated by political parties, (ii) party leaders maximize an organizational coalition prize on behalf of their cadre, and (iii) the post-election fiscal allocation rewards mayors aligned with the winning coalition through a discretionary transfer channel that party-cadre mayors can credibly access. Brazilian municipal politics from 2014 satisfies all three: the formal coalition string mechanically transmits the party leader’s strategic decision to all mayors of that party; the PT-PMDB-PSDB coalition system delivers credible post-election prize distribution; and the FPM / emendas mechanism is large, discretionary, and coalition-targeted.

Where the model fails. The French application of this section establishes the negative case. The mean-field framework does *not* recover a positive prize when (i) the endorsement signal is individual rather than party-mediated, (ii) the institutional rules attach a separate strategic motive (ballot qualification) to the signal, and (iii) the dominant determinant of the signal is ideological match between mayor and candidate rather than expected post-election prize allocation. French parrainage satisfies all three counterconditions: each parrainage is signed by name; the 500-signature threshold gives mayors of every party a tactical reason to sign for their preferred candidate regardless of winning prospects; and the alternative-

specific $\hat{\beta}_j$ estimates show clean ideological signatures that account for most of the cross-département variation in $\hat{\sigma}(\tau, j)$.

Implications for the literature. The structural-empirical literature on political endorsement has historically operated at one of two extremes: small-game full-enumeration formulations (Vitorino, 2012; Su, 2014) that handle a few dozen agents but are silent about scaling; or descriptive cross-country comparisons that sidestep structural identification. This paper’s contribution is in the middle: a model that scales to thousands of strategic agents under one institutional structure (Brazilian coalitions) and that explicitly delineates the institutional preconditions under which the scaling is meaningful (the French scope-condition test). The natural model extension that the French finding motivates — adding an explicit ballot-qualification motive or endorsement cost to the payoff specification — would restore the model’s applicability to individual sincere-signaling systems while preserving the mean-field tractability advantage that makes the Brazilian application possible. We leave that extension to future work.

9 Conclusion

The structural framework developed in this paper enables estimation of mayoral endorsement games at the scale of an entire country, a regime in which prior full-enumeration formulations (Su, 2014; Vitorino, 2012) are computationally infeasible. The mean-field equilibrium reduces the $(J+1)^I$ joint-profile state space to a J -dimensional fixed point on the candidate simplex, BLP-style logit inversion delivers identification, and Monte Carlo at Brazilian scale confirms $\sqrt{n}$ consistency with near-nominal coverage under PSD structural-delta standard errors.

We apply the framework to two contrasting institutional settings. In Brazil’s coalition-mediated endorsement system, using the 2014 cycle with a coalition-based proxy, we obtain estimates of the structural budget-prize parameter in the range $\hat{\alpha} \cdot \hat{B} \in [0.56, 0.65]$ ($t > 9$) across three bloc-augmented type partitions, with bloc fixed effects that trace the expected coalition pattern; a bloc-free state-only baseline yields 0.27, which we read as omitted-variable bias once bloc identity is the dominant within-state predictor of endorsement. A Brollo-style external anchor for B via parliamentary-amendment data returns a null result, which we attribute to the coarseness of the coalition-based proxy: PMDB and other Centrão mayors are mechanically labelled as aligned with the PT presidential candidate, but their actual access to discretionary federal money is famously coalition-independent. In France’s parrainages system, using both 2017 and 2022 cycles with the alternative-specific estimator on $\approx 9,500$ mayor sponsors per cycle and $J = 11$ –12 qualifying candidates, we recover ideologically-interpretable per-candidate covariate effects (Macron urban + peri-urban, Mélenchon dense urban, Pécresse rural) but the prize parameter mechanically col-

lapses to $\hat{\alpha} \cdot \hat{B} \approx -0.12$ because the equilibrium share equation cannot reconcile Macron’s small mayor-sponsorship share ($\approx 4\%$) with a positive prize. We interpret the contrast as a sharp scope condition: the prize-driven mean-field mechanism captures endorsement when the signal is mediated by political parties with prize-maximization incentives, but does not capture endorsement when the signal is individual, ideological, and tactically tied to candidate ballot-qualification.

Two natural follow-ups extend the present analysis. On the empirical side, a news-corpus-based extraction of real public endorsement statements in Brazil (Folha de São Paulo, O Globo, Estadão, Poder360) processed via a Portuguese named-entity-recognition pipeline would replace the coalition proxy with a finer mayor-level alignment indicator and permit revisiting the alignment-premium anchor for B , plus adding the 2018 and 2022 cycles to identify α and B separately via cross-cycle structural variation. On the methodological side, the French scope-condition finding motivates a model extension that adds an explicit endorsement cost or a ballot-qualification motive to the payoff specification — restoring the model’s applicability to individual sincere-signaling systems. Two further extensions of immediate interest are (i) a dynamic version of the game allowing for cycle-to-cycle learning, in the spirit of [Weintraub et al. \(2008\)](#) and [Adlakha et al. \(2015\)](#), and (ii) a network extension allowing for spatial spillovers in mayoral endorsement decisions, in the spirit of [Mele \(2017\)](#). All leave the methodological core of the present paper intact.

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A Appendix Tables and Figures

Appendix Table A1. Even More Regressions

	(1)	(2)	(3)
	First Outcome	Second Outcome	Third...
A. Top Panel			
Post Transition	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)
Mean	0.00	0.00	0.00
<i>N</i>	0,000,000	0,000,000	0,000,000
B. Bottom Panel			
Post Transition	0.0000*** (0.0000)	0.0000*** (0.0000)	0.0000*** (0.0000)
Mean	0.00	0.00	0.00
<i>N</i>	0,000,000	0,000,000	0,000,000

This table presents... * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

B Existence and Uniqueness of Mean-Field Equilibrium

This appendix proves the existence claim of Proposition 1 and provides a sufficient condition for uniqueness within the parameter region of empirical interest. Throughout, the structural parameter vector $\theta = (\beta, \alpha, B)$ is held fixed.

B.1 Setup and the share-implied best-response operator

For each share vector $s \in \mathring{\Delta}^J = \{s \in \Delta^J : s_j > 0 \forall j\}$ and each type $\tau \in \mathcal{T}$, the best-response choice probability is

$$\sigma(j \mid \tau; s, \theta) = \frac{\exp(\pi(j, s, \tau; \theta))}{\sum_k \exp(\pi(k, s, \tau; \theta))}, \quad (29)$$

where $\pi(j, s, \tau; \theta)$ is given by equation (3) of the main text. Define the share-implied best-response operator $\Phi : \mathring{\Delta}^J \rightarrow \mathring{\Delta}^J$ by

$$\Phi(s)_j = \sum_{\tau \in \mathcal{T}} f(\tau) w(\tau) \sigma(j \mid \tau; s, \theta), \quad j = 0, 1, \dots, J. \quad (30)$$

A mean-field equilibrium is a fixed point $s^* = \Phi(s^*)$. We work with $\mathring{\Delta}^J$ rather than the closed simplex Δ^J to keep the prize term B_j/s_j well-defined; the closure to the boundary is established at the end of the existence proof.

B.2 Existence

Proposition 2 (Existence; restatement of Proposition 1). *For every $\theta = (\beta, \alpha, B)$ with $\alpha > 0$, $B_j > 0$ for all $j > 0$, and finite β , the operator Φ admits a fixed point in $\mathring{\Delta}^J$.*

Proof. We restrict attention to a closed convex subset of $\mathring{\Delta}^J$ on which Φ is well-defined and continuous. Choose any $\bar{s} \in (0, 1/(J+1))$ small enough that the payoff $\pi(j^*, s, \tau; \theta)$, evaluated at $s_{j^*} = \bar{s}$, takes the largest value attained anywhere on the simplex — equivalently, $\alpha B_{j^*}/\bar{s}$ is large enough that the candidate-winner choice probability under that payoff exceeds $\bar{s}$. Concretely, define

$$\mathcal{D}_{\bar{s}} = \{s \in \Delta^J : s_j \geq \bar{s} \text{ for all } j = 0, \dots, J, \text{ and at most one } j \neq j^* \text{ has } s_j > 1/(J+1)\},$$

which contains a neighborhood of any candidate equilibrium with $s_{j^*}^*$ bounded away from zero. $\mathcal{D}_{\bar{s}}$ is compact, convex, and non-empty for $\bar{s}$ sufficiently small.

Self-mapping. On $\mathcal{D}_{\bar{s}}$ the prize term $B_{j^*}/s_{j^*} \leq B_{j^*}/\bar{s}$ is bounded, so $\sigma(j \mid \tau; s, \theta)$ is bounded away from 0 uniformly in $(\tau, s) \in \mathcal{T} \times \mathcal{D}_{\bar{s}}$. Aggregating over types gives $\Phi(s)_j \geq \sigma_{\min}(\bar{s}) > 0$ for every j and $s \in \mathcal{D}_{\bar{s}}$. By construction we choose $\bar{s} \leq \sigma_{\min}(\bar{s})$, ensuring $\Phi(\mathcal{D}_{\bar{s}}) \subseteq \mathcal{D}_{\bar{s}}$. Existence of such $\bar{s}$ follows because $\sigma_{\min}(\bar{s})$ is bounded below by the multinomial-logit lower envelope $1/(J+1+\exp(\overline{X'}\beta+\alpha B_{j^*}/\bar{s}))$, which approaches a positive

constant as $\bar{s} \rightarrow 0^+$ from the dominant-prize side, yielding a fixed point of the inequality $\bar{s} \leq \sigma_{\min}(\bar{s})$ at small $\bar{s}$.

Continuity. $\pi(j, s, \tau; \theta)$ is continuous in s on $\mathcal{D}_{\bar{s}}$ for each j that is locally a winner; the winner indicator $\mathbf{1}\{j \in W(s)\}$ is locally constant on the open set where s_{j^*} is the strict argmax over $\{1, \dots, J\}$. Restricting attention to that open set (which is non-empty by the dominant-prize choice of $\bar{s}$) gives Φ continuous as a composition of continuous functions. The complementary tie-set is measure-zero and we adopt the convention of randomizing uniformly among co-winners, which preserves continuity.

Brouwer. $\Phi : \mathcal{D}_{\bar{s}} \rightarrow \mathcal{D}_{\bar{s}}$ is a continuous self-map of a compact convex set, so a fixed point $s^* \in \mathcal{D}_{\bar{s}}$ exists. $\square$

B.3 Local uniqueness via numerical contraction

Existence does not preclude multiple equilibria. We do not provide a general analytic sufficient condition for uniqueness because the share-implied best-response operator Φ is not globally a contraction: the elasticity of the type-conditional choice probability $\sigma(j^* \mid \tau; s)$ with respect to s_{j^*} ,

$$\frac{\partial \log \sigma(j^* \mid \tau; s)}{\partial \log s_{j^*}} = - \frac{\alpha B_{j^*}}{s_{j^*}} \cdot (1 - \sigma(j^* \mid \tau; s)), \quad (31)$$

diverges in absolute value as $s_{j^*} \rightarrow 0$. The operator can be a contraction only on a sublevel set bounded away from zero, and only when $\alpha B_{j^*}/s_{j^*}^2$ is small enough that the projected one-dimensional dynamic on the subspace $\{s : s_0 = s_j \text{ for all } j \neq j^*\}$ has $|\Phi'_{j^*}(s_{j^*})| < 1$. A bound of $|\Phi'_{j^*}(s_{j^*})| \leq \alpha B_{j^*}/(4s_{j^*}^2)$ follows from (31) together with $\sigma(1 - \sigma) \leq 1/4$, so contraction holds on $\{s_{j^*} \geq \underline{s}\}$ whenever $\alpha B_{j^*} < 4\underline{s}^2$. Larger αB_{j^*} values — the empirically interesting region — accordingly do *not* satisfy this analytic sufficient condition.

We instead verify uniqueness numerically within the parameter region of interest by iterating Φ from a grid of starting points and checking that all converge to the same equilibrium. Section 6.1 reports the result for the Monte Carlo DGP: with $\alpha = 1.5$, $B = (1.0, 0.4, 0.2)$, the empirical equilibrium at $s_{j^*}^* \approx 0.708$ is recovered from 20 random initial points in $\Delta_{0.01}^J$ in every case. (Three coordination equilibria coexist globally at this θ — one per candidate-as-winner — but only the $j^* = 1$ equilibrium is stable in the sense that Φ 's spectral radius at the fixed point is below one. The analytical derivation of this spectral-radius condition is the content of Bodoh-Creed (2013) §3.) The empirical resolution rule for ambient multiplicity is discussed in Section 3.4.

B.4 Approximation of the finite- I Bayesian game

The mean-field equilibrium is the limit of finite- I equilibria as $I \rightarrow \infty$. The relevant static-large-game approximation is Bodoh-Creed (2013): for any mean-field equilibrium (σ^*, s^*)

and any $\epsilon > 0$, there exists $I_0(\epsilon)$ such that for all $I \geq I_0$, (σ^*, s^*) is an ϵ -Bayesian equilibrium of the corresponding finite- I game (i.e., no mayor can profitably deviate by more than ϵ). The bound takes the form $\epsilon = O(1/\sqrt{I})$ under regularity conditions that are satisfied on the lower-bounded simplex restriction $\mathcal{D}_{\bar{s}}$ used for existence above (the prize term B_{j^*}/s_{j^*} is finite and Lipschitz in s on $\mathcal{D}_{\bar{s}}$). For Brazil's $I = 5,570$, the implied bound is approximately 0.013, an order of magnitude smaller than typical estimation standard errors and therefore innocuous for the empirical exercise.

C Appendix Code: Original Full-Enumeration GAMS Implementation

Note: This appendix retains the original GAMS implementation that estimates the model under full enumeration of joint endorsement profiles. It is the code referenced in the runtime comparison of Section 6.5 as the small-game baseline against which the mean-field formulation is contrasted. The current preferred implementation is in Python and lives at `code/python/mc_v2/` with reproducibility at empirical scale; the Python re-implementation of this exact GAMS algorithm (for verification of the Section 6.5 runtime numbers) is at `code/python/mc/`.

C.1 GAMS code for the original full-enumeration Monte Carlo

Listing 1. GAMS Example

```
$title  BRAZIL MAYORAL ENDORSEMENT ON PRESIDENTIAL ELECTIONS

$ontext
----- +
Last      update      : 06/21/2024          +
Contact Pengzhen Li: pli26@vols.utk.edu      +
----- +

$ondollar
$Oneolcom
$offlisting

Sets
I      'number of mayors; 3 mayors are assumed' /i1*i3/
J      'number of candidates; 2 candidates are assumed and j0 means no
      endorsement' /j0*j2/
J_0(J) 'subset of J that exclude j0' /j1*j2/
t      'time periods' /t1*t1000/
n      'number of cov in X; 2 covariates are assumed in the public information'
      /n1*n2/ ;
alias (J, J1, J2, J3) 'create alternative nameS for a set J'

Parameter
X(I,n) 'simulated covariates'
      /i1.n1 -4.378051, i1.n2 -14.265966,
      i2.n1 -50.582518, i2.n2 -24.406213,
      i3.n1 -9.810775, i3.n2 4.967237/
Y_count(I,J) 'tabulation of endorsement outcomes over 1000 periods'
$ontext
/i1.j0 9, i1.j1 486, i1.j2 505,
i2.j0 215, i2.j1 210, i2.j2 575,
i3.j0 531, i3.j1 272, i3.j2 197/
$offtext
```

```

/i1.j0 10.29168, i1.j1 496.1649, i1.j2 493.5434,
i2.j0 216.56027, i2.j1 233.5908, i2.j2 549.8489,
i3.j0 518.44169, i3.j1 279.4408, i3.j2 202.1175/

```

```

winmayor(J1,J2,J3,I)

```

```

/j0.j0.j0.i1 0, j0.j0.j0.i2 0, j0.j0.j0.i3 0,
j1.j0.j0.i1 1, j1.j0.j0.i2 0, j1.j0.j0.i3 0,
j2.j0.j0.i1 1, j2.j0.j0.i2 0, j2.j0.j0.i3 0,
j0.j1.j0.i1 0, j0.j1.j0.i2 1, j0.j1.j0.i3 0,
j1.j1.j0.i1 1, j1.j1.j0.i2 1, j1.j1.j0.i3 0,
j2.j1.j0.i1 1, j2.j1.j0.i2 1, j2.j1.j0.i3 0,
j0.j2.j0.i1 0, j0.j2.j0.i2 1, j0.j2.j0.i3 0,
j1.j2.j0.i1 1, j1.j2.j0.i2 1, j1.j2.j0.i3 0,
j2.j2.j0.i1 1, j2.j2.j0.i2 1, j2.j2.j0.i3 0,
j0.j0.j1.i1 0, j0.j0.j1.i2 0, j0.j0.j1.i3 1,
j1.j0.j1.i1 1, j1.j0.j1.i2 0, j1.j0.j1.i3 1,
j2.j0.j1.i1 1, j2.j0.j1.i2 0, j2.j0.j1.i3 1,
j0.j1.j1.i1 0, j0.j1.j1.i2 1, j0.j1.j1.i3 1,
j1.j1.j1.i1 1, j1.j1.j1.i2 1, j1.j1.j1.i3 1,
j2.j1.j1.i1 0, j2.j1.j1.i2 1, j2.j1.j1.i3 1,
j0.j2.j1.i1 0, j0.j2.j1.i2 1, j0.j2.j1.i3 1,
j1.j2.j1.i1 1, j1.j2.j1.i2 0, j1.j2.j1.i3 1,
j2.j2.j1.i1 1, j2.j2.j1.i2 1, j2.j2.j1.i3 0,
j0.j0.j2.i1 0, j0.j0.j2.i2 0, j0.j0.j2.i3 1,
j1.j0.j2.i1 1, j1.j0.j2.i2 0, j1.j0.j2.i3 1,
j2.j0.j2.i1 1, j2.j0.j2.i2 0, j2.j0.j2.i3 1,
j0.j1.j2.i1 0, j0.j1.j2.i2 1, j0.j1.j2.i3 1,
j1.j1.j2.i1 1, j1.j1.j2.i2 1, j1.j1.j2.i3 0,
j2.j1.j2.i1 1, j2.j1.j2.i2 0, j2.j1.j2.i3 1,
j0.j2.j2.i1 0, j0.j2.j2.i2 1, j0.j2.j2.i3 1,
j1.j2.j2.i1 0, j1.j2.j2.i2 1, j1.j2.j2.i3 1,
j2.j2.j2.i1 1, j2.j2.j2.i2 1, j2.j2.j2.i3 1/

```

```

winmayor_endorse1(J1,J2,J3,I)

```

```

/j0.j0.j0.i1 0, j0.j0.j0.i2 0, j0.j0.j0.i3 0,
j1.j0.j0.i1 1, j1.j0.j0.i2 0, j1.j0.j0.i3 0,
j2.j0.j0.i1 0, j2.j0.j0.i2 0, j2.j0.j0.i3 0,
j0.j1.j0.i1 0, j0.j1.j0.i2 1, j0.j1.j0.i3 0,
j1.j1.j0.i1 1, j1.j1.j0.i2 1, j1.j1.j0.i3 0,
j2.j1.j0.i1 0, j2.j1.j0.i2 1, j2.j1.j0.i3 0,
j0.j2.j0.i1 0, j0.j2.j0.i2 0, j0.j2.j0.i3 0,
j1.j2.j0.i1 1, j1.j2.j0.i2 0, j1.j2.j0.i3 0,
j2.j2.j0.i1 0, j2.j2.j0.i2 0, j2.j2.j0.i3 0,
j0.j0.j1.i1 0, j0.j0.j1.i2 0, j0.j0.j1.i3 1,
j1.j0.j1.i1 1, j1.j0.j1.i2 0, j1.j0.j1.i3 1,
j2.j0.j1.i1 0, j2.j0.j1.i2 0, j2.j0.j1.i3 1,
j0.j1.j1.i1 0, j0.j1.j1.i2 1, j0.j1.j1.i3 1,

```

```

j1.j1.j1.i1 1, j1.j1.j1.i2 1, j1.j1.j1.i3 1,
j2.j1.j1.i1 0, j2.j1.j1.i2 1, j2.j1.j1.i3 1,
j0.j2.j1.i1 0, j0.j2.j1.i2 0, j0.j2.j1.i3 1,
j1.j2.j1.i1 1, j1.j2.j1.i2 0, j1.j2.j1.i3 1,
j2.j2.j1.i1 0, j2.j2.j1.i2 0, j2.j2.j1.i3 0,
j0.j0.j2.i1 0, j0.j0.j2.i2 0, j0.j0.j2.i3 0,
j1.j0.j2.i1 1, j1.j0.j2.i2 0, j1.j0.j2.i3 0,
j2.j0.j2.i1 0, j2.j0.j2.i2 0, j2.j0.j2.i3 0,
j0.j1.j2.i1 0, j0.j1.j2.i2 1, j0.j1.j2.i3 0,
j1.j1.j2.i1 1, j1.j1.j2.i2 1, j1.j1.j2.i3 0,
j2.j1.j2.i1 0, j2.j1.j2.i2 0, j2.j1.j2.i3 0,
j0.j2.j2.i1 0, j0.j2.j2.i2 0, j0.j2.j2.i3 0,
j1.j2.j2.i1 0, j1.j2.j2.i2 0, j1.j2.j2.i3 0,
j2.j2.j2.i1 0, j2.j2.j2.i2 0, j2.j2.j2.i3 0/

```

```
winmayor_endorse2(J1,J2,J3,I)
```

```
winmayor_total(J1,J2,J3)
```

```
wincandidate(J,J1,J2,J3)
```

```

/j1.j0.j0.j0 0, j2.j0.j0.j0 0, j0.j0.j0.j0 0,
j1.j1.j0.j0 1, j2.j1.j0.j0 0, j0.j1.j0.j0 0,
j1.j2.j0.j0 0, j2.j2.j0.j0 1, j0.j2.j0.j0 0,
j1.j0.j1.j0 1, j2.j0.j1.j0 0, j0.j0.j1.j0 0,
j1.j1.j1.j0 1, j2.j1.j1.j0 0, j0.j1.j1.j0 0,
j1.j2.j1.j0 1, j2.j2.j1.j0 1, j0.j2.j1.j0 0,
j1.j0.j2.j0 0, j2.j0.j2.j0 1, j0.j0.j2.j0 0,
j1.j1.j2.j0 1, j2.j1.j2.j0 1, j0.j1.j2.j0 0,
j1.j2.j2.j0 0, j2.j2.j2.j0 1, j0.j2.j2.j0 0,
j1.j0.j0.j1 1, j2.j0.j0.j1 0, j0.j0.j0.j1 0,
j1.j1.j0.j1 1, j2.j1.j0.j1 0, j0.j1.j0.j1 0,
j1.j2.j0.j1 1, j2.j2.j0.j1 1, j0.j2.j0.j1 0,
j1.j0.j1.j1 1, j2.j0.j1.j1 0, j0.j0.j1.j1 0,
j1.j1.j1.j1 1, j2.j1.j1.j1 0, j0.j1.j1.j1 0,
j1.j2.j1.j1 1, j2.j2.j1.j1 0, j0.j2.j1.j1 0,
j1.j0.j2.j1 1, j2.j0.j2.j1 1, j0.j0.j2.j1 0,
j1.j1.j2.j1 1, j2.j1.j2.j1 0, j0.j1.j2.j1 0,
j1.j2.j2.j1 0, j2.j2.j2.j1 1, j0.j2.j2.j1 0,
j1.j0.j0.j2 0, j2.j0.j0.j2 1, j0.j0.j0.j2 0,
j1.j1.j0.j2 1, j2.j1.j0.j2 1, j0.j1.j0.j2 0,
j1.j2.j0.j2 0, j2.j2.j0.j2 1, j0.j2.j0.j2 0,
j1.j0.j1.j2 1, j2.j0.j1.j2 1, j0.j0.j1.j2 0,
j1.j1.j1.j2 1, j2.j1.j1.j2 0, j0.j1.j1.j2 0,
j1.j2.j1.j2 0, j2.j2.j1.j2 1, j0.j2.j1.j2 0,
j1.j0.j2.j2 0, j2.j0.j2.j2 1, j0.j0.j2.j2 0,
j1.j1.j2.j2 0, j2.j1.j2.j2 1, j0.j1.j2.j2 0,
j1.j2.j2.j2 0, j2.j2.j2.j2 1, j0.j2.j2.j2 0/

```

```
wincandidate_total(J1,J2,J3);
```

```

winmayor_total(J1,J2,J3) = sum(I,winmayor(J1,J2,J3,I));
winmayor_total('j0','j0','j0') = 1000000000000000000000000;
wincandidate_total(J1,J2,J3) = sum(J,wincandidate(J,J1,J2,J3));
wincandidate_total('j0','j0','j0') = 1000000000000000000000000;
winmayor_endorse2(J1,J2,J3,I) = winmayor(J1,J2,J3,I) - winmayor_endorse1(J1,
    J2,J3,I);

display winmayor,winmayor_total;

positive variable
sigma(I,J) 'probability matrix'
;
Equation
sigmaI(I);
sigmaI(I).. SUM(J,sigma(I,J)) =e= 1;
variable
beta(n)
sigma_minus(J1,J2,J3,I)
B(I) ;
sigma_minus.fx(J1,J2,J3,I)$(NOT winmayor(J1,J2,J3,I))=0;
Equation
sigma_minus1(J1,J2,J3)
sigma_minus2(J1,J2,J3)
sigma_minus3(J1,J2,J3);
sigma_minus1(J1,J2,J3)$winmayor(J1,J2,J3,'i1').. sigma_minus(J1,J2,J3,'i1')=
    E= sigma('i2',J2)*sigma('i3',J3)/wincandidate_total(J1,J2,J3);
sigma_minus2(J1,J2,J3)$winmayor(J1,J2,J3,'i2').. sigma_minus(J1,J2,J3,'i2')=
    E= sigma('i1',J1)*sigma('i3',J3)/wincandidate_total(J1,J2,J3);
sigma_minus3(J1,J2,J3)$winmayor(J1,J2,J3,'i3').. sigma_minus(J1,J2,J3,'i3')=
    E= sigma('i1',J1)*sigma('i2',J2)/wincandidate_total(J1,J2,J3);

variable
logsigma(I,J)
;
equation
logsigmaIJ(I,J);
logsigmaIJ(I,J).. sigma(I,'j0')*exp(logsigma(I,J))=E=sigma(I,J);

VARIABLE
pi_lower_b(J1,J2,J3,I)
pi_lower_xbeta(J1,J2,J3,I)
pi_lower(J1,J2,J3,I)
pi_up(I,J);

Equation
pi_lower_bJI(J1,J2,J3,I)
pi_lower_xbetaJI(J1,J2,J3,I)

```



```

pi_lowerJI(J1,J2,J3,I)
pi_up0(I)
pi_up1(I)
pi_up2(I);

pi_lower_bJI(J1,J2,J3,I).. pi_lower_b(J1,J2,J3,I) =E= B(I)*(1/winmayor_total
    (J1,J2,J3))*winmayor(J1,J2,J3,I);
pi_lower_xbetaJI(J1,J2,J3,I).. pi_lower_xbeta(J1,J2,J3,I) =E= sum(n,X(I,n)*
    beta(n))*winmayor(J1,J2,J3,I);
pi_lowerJI(J1,J2,J3,I).. pi_lower(J1,J2,J3,I)=E= pi_lower_b(J1,J2,J3,I) +
    pi_lower_xbeta(J1,J2,J3,I) ;
pi_up1(I).. pi_up(I,'j1') =E= SUM((J1,J2,J3),pi_lower(J1,J2,J3,I)*
    sigma_minus(J1,J2,J3,I)$winmayor_endorse1(J1,J2,J3,I));
pi_up2(I).. pi_up(I,'j2') =E= SUM((J1,J2,J3),pi_lower(J1,J2,J3,I)*
    sigma_minus(J1,J2,J3,I)$winmayor_endorse2(J1,J2,J3,I));
pi_up0(I).. pi_up(I,'j0') =E=0;

*$ontext
equation
gama(I,J);
gama(I,J_0).. pi_up(I,J_0)=E= logsigma(I,J_0);
*$offtext

VARIABLE
sigmaE(I,J)
sumsigmaE(I);

equation
sigmaEIJ(I,J)
sumsigmaEI(I);
sumsigmaEI(I).. sumsigmaE(I) =E= sum(J,exp(pi_up(I,J)));
sigmaEIJ(I,J).. sumsigmaE(I)*sigmaE(I,J)=E= exp(pi_up(I,J));

VARIABLE
MAXLIK;
equation
MAXLIKE;
MAXLIKE.. MAXLIK =e= sum((I,J),log(sigmaE(I,J)+0.000001)*Y_count(I,J));

Model Endorsement "non-linear" /all/;

option NLP = lindo;
option optcr = 0.000001;

solve Endorsement using NLP MAX MAXLIK;

execute_unload "endorse_baron.gdx"

```

```
sigma, sigma_minus, winmayor, winmayor_endorse1, winmayor_endorse2,
    winmayor_total, wincandidate, wincandidate_total,
X, logsigma, pi_lower_b, beta, pi_lower_xbeta, pi_lower, pi_up, B, MAXLIK;
$ontext
*$ontext
$onecho > convertD.opt
dumpGDX h.gdx
dictmap d.gdx
$offecho
option solver=convertD; Endorsement.optFile = 1;
solve Endorsement using NLP MAX MAXLIK;
$offtext
```